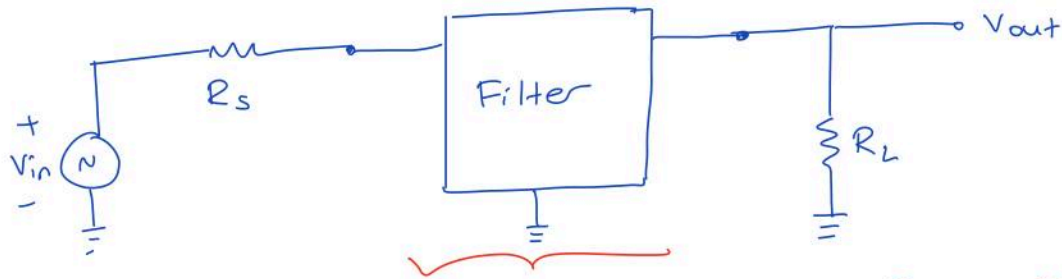


- filters provide frequency selectivity
- tuned circuits also provide selectivity
- we want more sophisticated filters, with higher Q

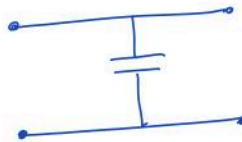


Goal: maintain wanted frequencies
eliminate unwanted frequencies

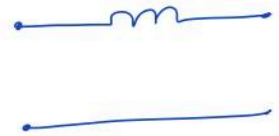
- Typically, filters are made of L & C.

Low pass filters:

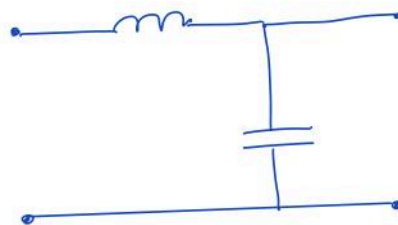
First-order:



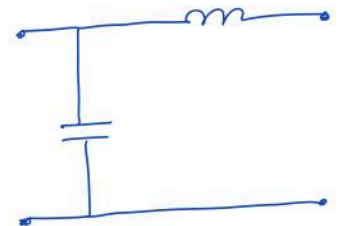
OR



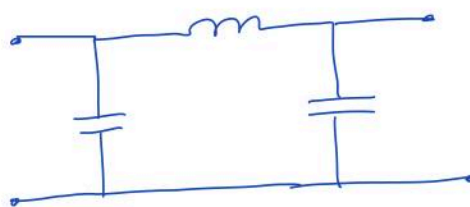
Second-order:



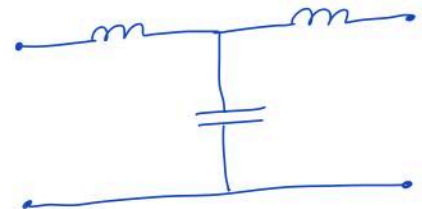
OR



Third-order:

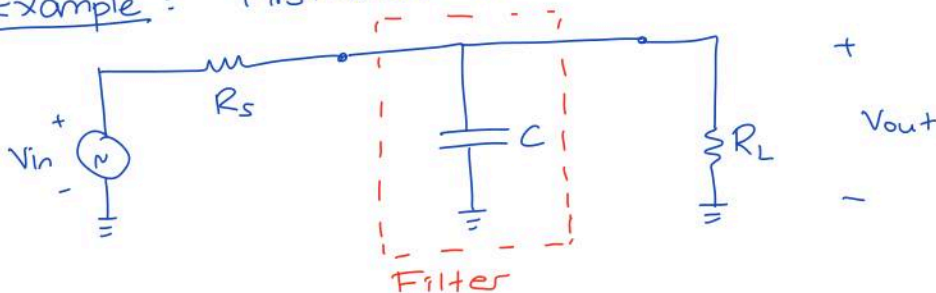


OR



closer
to
ideal

Example: First-order LPF



$$H_1(\omega) = \frac{V_{out}}{V_{in}} = \frac{R_L \parallel \frac{1}{j\omega C}}{R_S + (R_L \parallel \frac{1}{j\omega C})} = \frac{\frac{R_L}{1+j\omega R_L C}}{R_S + \frac{R_L}{1+j\omega R_L C}} \quad (2)$$

$$= \frac{R_L}{(R_L + R_S) + j\omega R_L R_S C} = \left(\frac{R_L}{R_L + R_S} \right) \cdot \frac{1}{1 + j\omega \frac{R_L R_S}{R_L + R_S} \cdot C}$$

3-dB cut-off freq: $|H_1(\omega_c)| = \frac{1}{\sqrt{2}} \max |H_1(\omega)|$

$$\max |H_1(\omega)| = \frac{R_L}{R_L + R_S}$$

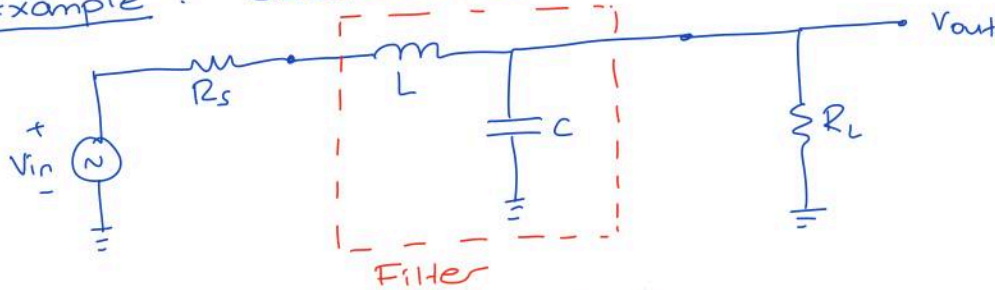
$$\omega_c \frac{R_L R_S}{R_L + R_S} \cdot C = 1 \Rightarrow \omega_c = \frac{1}{\underbrace{\frac{R_L R_S}{R_L + R_S}}_{R_{eq} = R_L \parallel R_S} \cdot C}$$

Then,

$$H_1(\omega) = \frac{R_L}{R_L + R_S} \cdot \frac{1}{1 + j \frac{\omega}{\omega_c}}$$

* Assume $R_L = R_S$: $H_1(\omega) = \frac{1}{2} \frac{1}{(1 + j \frac{\omega}{\omega_c})}$

Example: Second-order LPF



$$H_2(\omega) = \frac{V_{out}}{V_{in}} = \frac{R_L \parallel \frac{1}{j\omega C}}{R_S + j\omega L + (R_L \parallel \frac{1}{j\omega C})} = \frac{\frac{R_L}{1 + j\omega R_L C}}{R_S + j\omega L + \frac{R_L}{1 + j\omega R_L C}}$$

$$= \frac{R_L}{(R_S + j\omega L)(1 + j\omega R_L C) + R_L} = \frac{R_L}{R_S + R_L - \omega^2 R_L L C + j\omega(L + R_S R_L C)}$$

$$= \frac{R_L}{R_S + R_L} \cdot \frac{1}{1 - \frac{\omega^2 R_L L C}{R_S + R_L} + j\omega \frac{(L + R_S R_L C)}{R_S + R_L}}$$

* Choose $R_L = R_S = R$ (filter design choice #1)

(3)

$$H_2(\omega) = \frac{1}{2} \cdot \frac{1}{1 - \frac{\omega^2 LC}{2} + j\omega \left(\frac{L}{2R} + \frac{RC}{2} \right)}$$

* Choose $R^2 = \frac{L}{C}$ (filter design choice #2)

$$H_2(\omega) = \frac{1}{2} \cdot \frac{1}{1 - \frac{\omega^2 LC}{2} + j\omega \sqrt{LC}}$$

$$* |H_2(\omega)| = \frac{1}{2} \cdot \frac{1}{\sqrt{\left(1 - \frac{\omega^2 LC}{2}\right)^2 + \omega^2 LC}} = \frac{1}{2} \cdot \frac{1}{\sqrt{1 + \frac{\omega^4 L^2 C^2}{4}}}$$

* at $\omega=0$, $\max |H_2(\omega)| = |H_2(0)| = \frac{1}{2}$

* 3-dB cut-off freq: $|H_2(\omega_c)| = \frac{\max |H_2(\omega)|}{\sqrt{2}}$

$$\frac{\omega_c^4 L^2 C^2}{4} = 1 \Rightarrow \omega_c = \sqrt{\frac{2}{LC}}$$

* Then,

$$H_2(\omega) = \frac{1}{2} \cdot \frac{1}{1 - \left(\frac{\omega}{\omega_c}\right)^2 + j\sqrt{2} \left(\frac{\omega}{\omega_c}\right)}$$

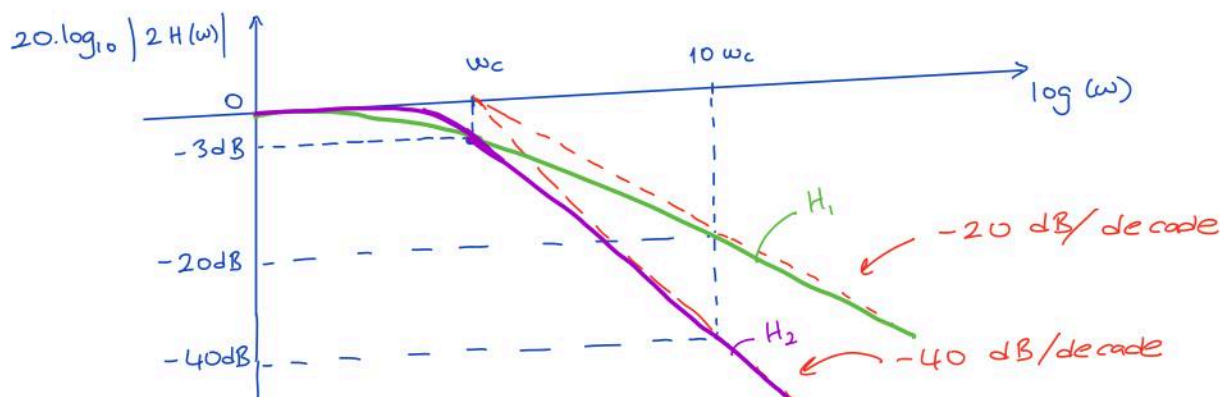
Compare 1st and 2nd order LPFs :

$$H_1(\omega) = \frac{1}{2 \left[1 + j \frac{\omega}{\omega_c} \right]}, \quad \text{if } \omega \gg \omega_c, |H_1(\omega)| \approx \frac{\omega_c}{2\omega}$$

-20 dB/decade

$$H_2(\omega) = \frac{1}{2 \left[1 - \left(\frac{\omega}{\omega_c}\right)^2 + j\sqrt{2} \left(\frac{\omega}{\omega_c}\right) \right]}, \quad \text{if } \omega \gg \omega_c, |H_2(\omega)| \approx \frac{1}{2} \cdot \left(\frac{\omega_c}{\omega}\right)^2$$

-40 dB/decade

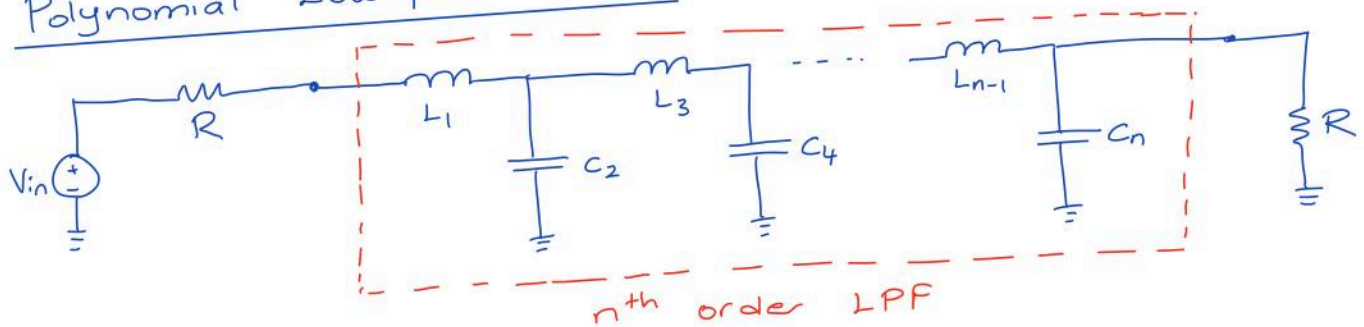


- * 2nd order is superior because
- (1) it suppresses high frequencies better
 - (2) it preserves low frequencies better.

Polynomial Filters

- transfer function is a polynomial (Chebyshev, elliptic, etc... We will use Butterworth filters)
- element values are obtained from lookup tables.
- Circuit morphology resembles a ladder.

Polynomial Low-pass Filters :



Remember :

* For inductor, $Z_L = j\omega L$

$\lim_{\omega \rightarrow 0} Z_L = 0$
 inductor acts like short circuit at DC

$\lim_{\omega \rightarrow \infty} Z_L = \infty$

inductor acts like open circuit at VERY high freq.

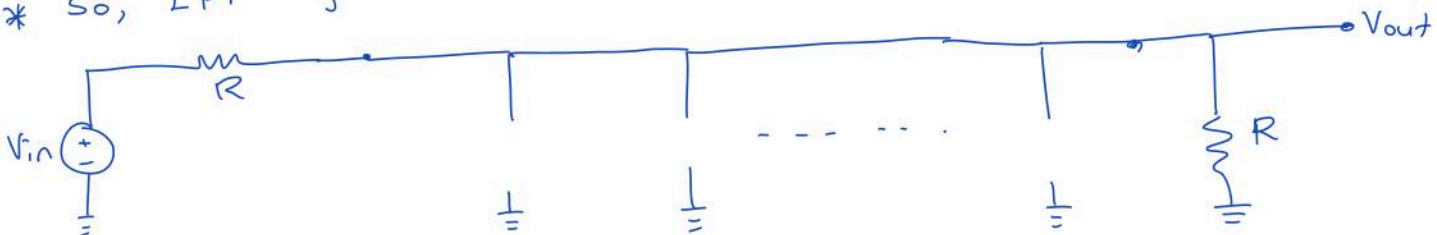
* For capacitor, $Z_C = \frac{1}{j\omega C}$

$\lim_{\omega \rightarrow 0} Z_C = \infty$
 capacitor acts like open circuit at DC

$\lim_{\omega \rightarrow \infty} Z_C = 0$

capacitor acts like short circuit at VERY high freq.

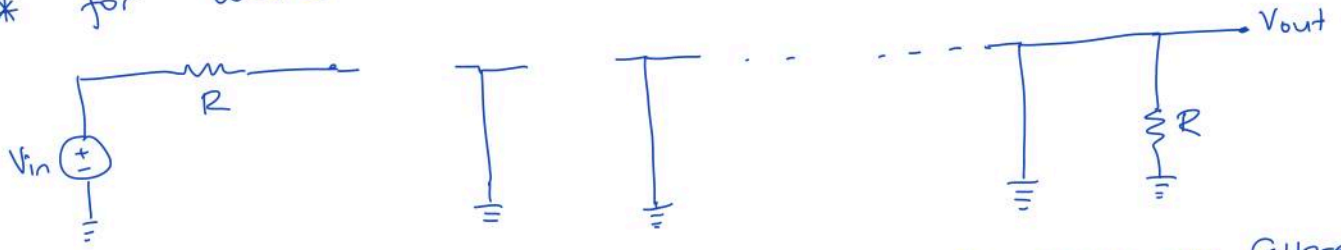
* So, LPF for $\omega \rightarrow 0$



$$V_{out} = \frac{R}{R+R} \cdot V_{in} = \frac{V_{in}}{2} \quad (\text{Not filtered})$$

(5)

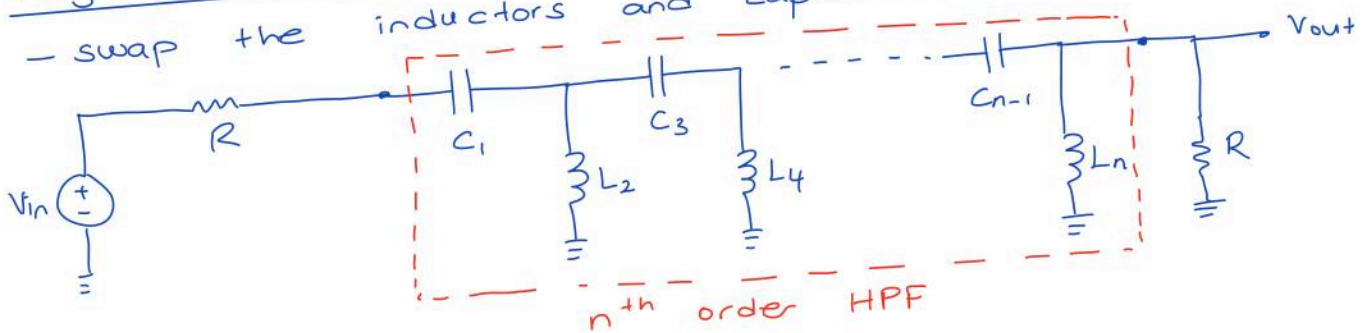
* for $\omega \rightarrow \infty$



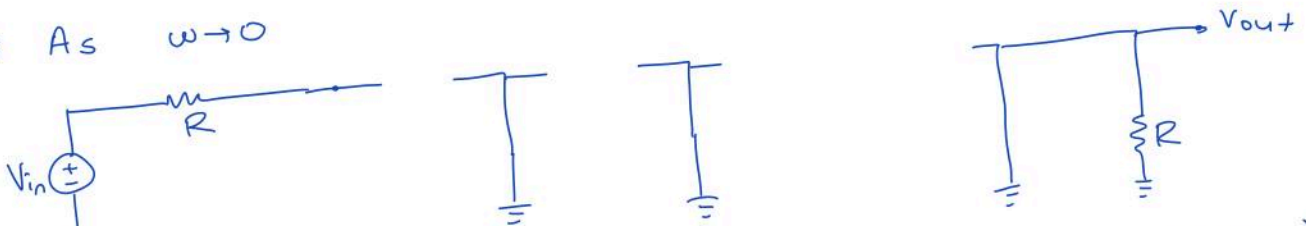
$V_{out} = 0$ (VERY high frequencies are filtered)

Polynomial High-pass Filters:

- swap the inductors and capacitors

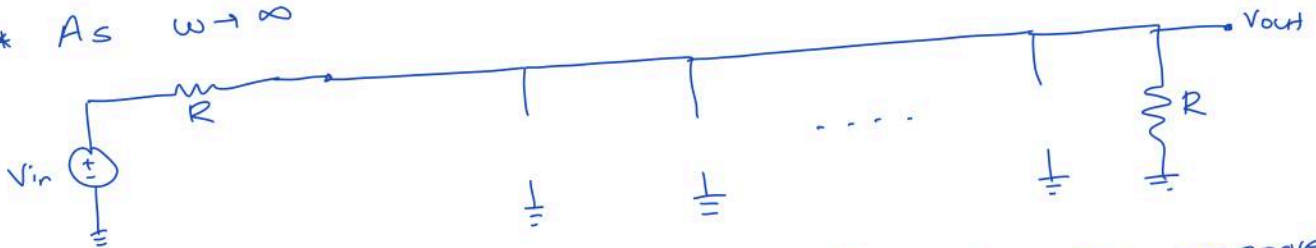


* As $\omega \rightarrow 0$



$V_{out} = 0$ (low frequencies are filtered)

* As $\omega \rightarrow \infty$



$V_{out} = \frac{V_{in}}{2}$ (high frequencies are preserved)

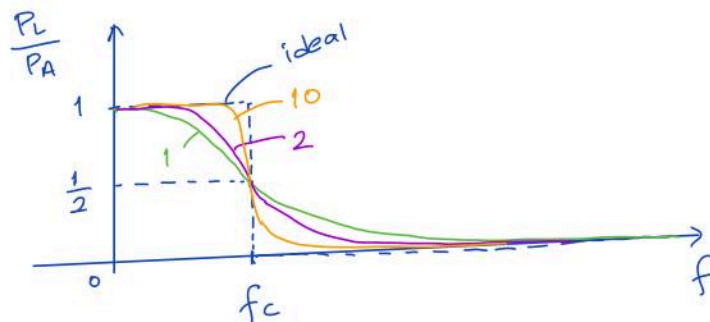
Butterworth Filters

Design goal is to have a specific power transfer function:

$$\frac{P_L}{P_A} = \frac{1}{1 + \left(\frac{f}{f_c}\right)^{2n}} \quad (\text{for Butterworth LPF})$$

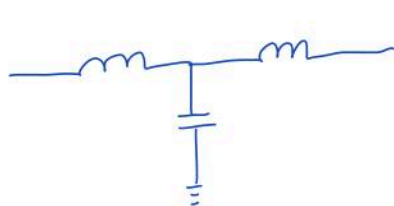
$\frac{P_L}{P_A}$: power delivered to load
 P_A : power available at the source: $P_A = \frac{|V_{in}|^2}{8R_s}$
 n : number of elements
 f_c : 3-dB cut-off freq.

* As n increases, power transfer function approaches an ideal LPF.

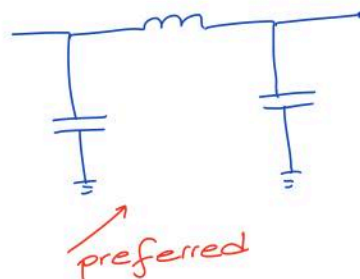


$$10 \log_{10} \left(\frac{1}{2} \right) = -3 \text{ dB}$$

- n can be even or odd
- ladder filter can start with a series or parallel component.
- typically, filters with fewer inductors are preferred.



OR



- Element values are provided in tables:

Butterworth filter coefficients

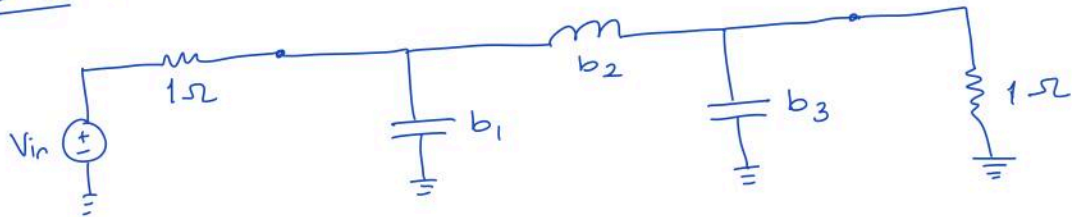
n	b_1	b_2	b_3	b_4
1	2			
2	1.414	1.414		
3	1	2	1	
4	0.7654	1.8478	1.8478	0.7654

more generally: $b_i = 2 \cdot \sin\left(\frac{(2i-1)\pi}{2n}\right)$ (7)

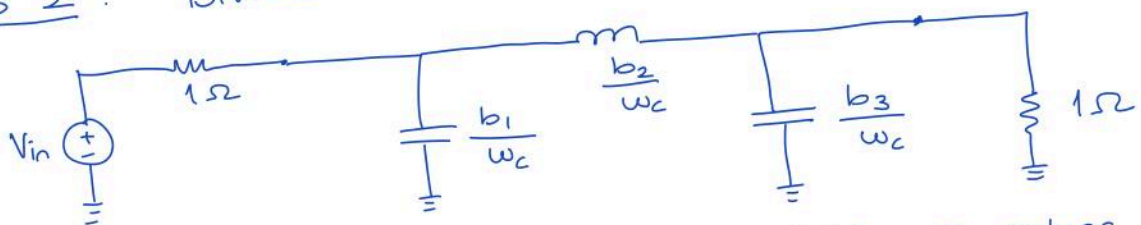
** These are normalized values for $\omega_c = 1 \text{ rad/s}$, **
and $R_s = R_L = R = 1 \Omega$.

Example: Suppose we need a LPF,
with $f_c = 20 \text{ kHz}$, $R = R_s = R_L = 500 \Omega$, $n = 3$.

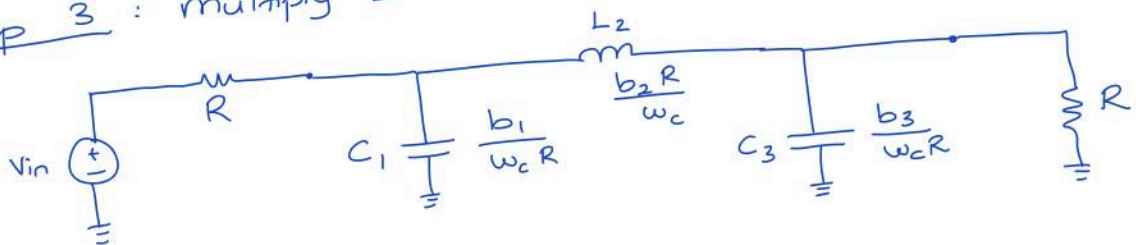
Step 1: Start with filter coefficients from the table:



Step 2: Divide L & C values by frequency:



Step 3: multiply L values with R, divide C values by R:



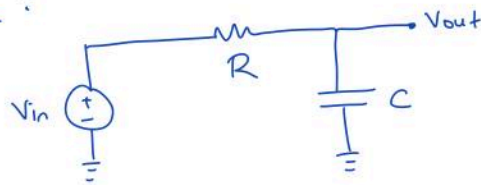
So,

$$C_1 = \frac{b_1}{\omega_c R} = \frac{b_1}{2\pi f_c \cdot R} = \frac{1}{2\pi \cdot 20 \cdot 10^3 \cdot 500} = 15.9 \text{ nF}$$

$$L_2 = \frac{b_2 R}{\omega_c} = \frac{2 \cdot 500}{2\pi \cdot 20 \cdot 10^3} = 7.96 \text{ mH}$$

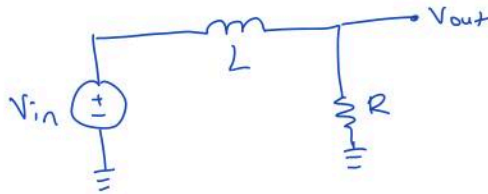
$$C_3 = \frac{b_3}{\omega_c R} = 15.9 \text{ nF}$$

Side note :



$$\omega_c = \frac{1}{RC}$$

$$C = \frac{1}{\omega_c R}$$



$$\omega_c = \frac{R}{L}$$

$$L = \frac{R}{\omega_c}$$

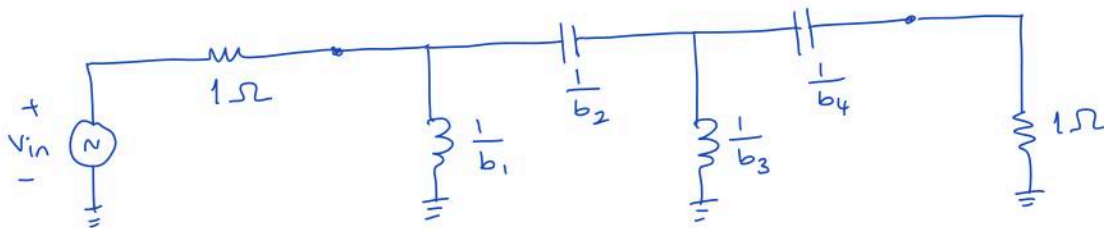
Butterworth HPF :

$$\frac{P_L}{P_A} = 1 - \underbrace{\left(\frac{1}{1 + \left(\frac{f}{f_c} \right)^{2n}} \right)}_{\text{LPF}} = \frac{\left(\frac{f}{f_c} \right)^{2n}}{1 + \left(\frac{f}{f_c} \right)^{2n}}$$

* Same filter coefficients as LPF, with one difference in the steps: we take the inverse of the coefficients:

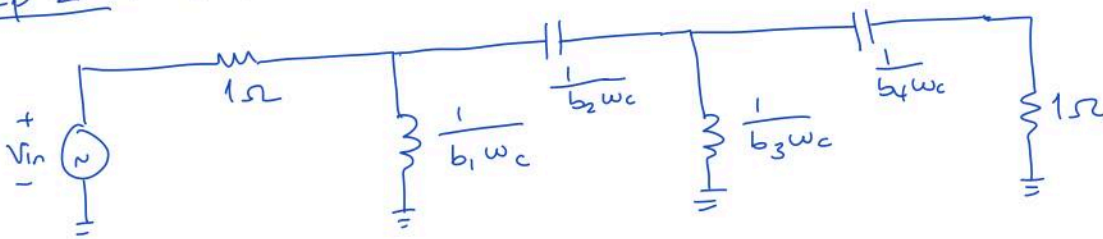
Example: Suppose we need a HPF with $f_c = 10 \text{ MHz}$, $R_s = R_L = R = 50 \Omega$, $n = 4$

Step 1: start with the inverse of the filter coefficients.
 $b_1 = 0.7654$, $b_2 = 1.8478$, $b_3 = 1.8478$, $b_4 = 0.7654$



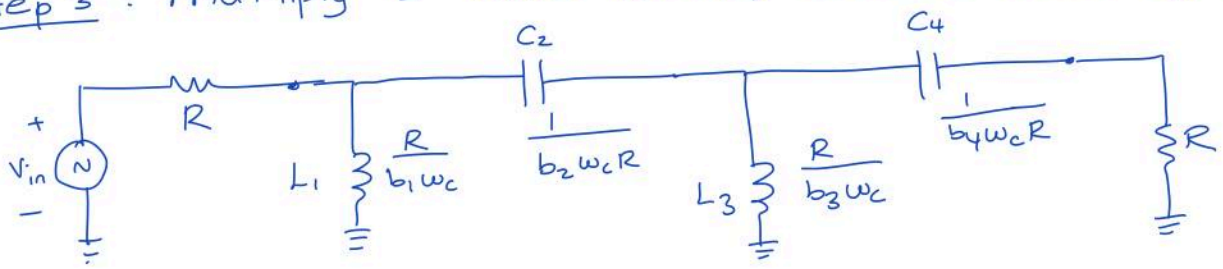
$$\omega_c = 1 \text{ rps}$$

Step 2: Divide L & C values by frequency:



$$\omega_c = 2\pi \cdot 10 \cdot 10^6 \text{ rad/s}$$

Step 3 : multiply L values with R, divide C values by R: (9)



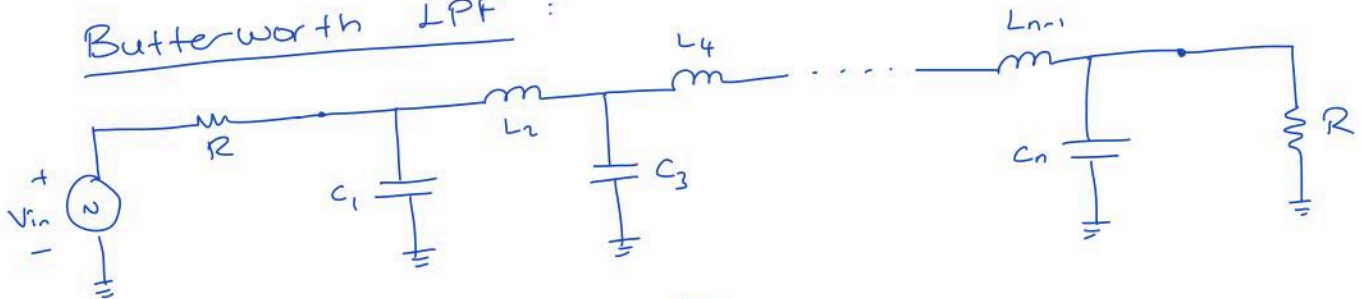
$$\text{So, } L_1 = \frac{R}{b_1 \omega_c} = \frac{50}{0.7654 \cdot 2\pi \cdot 10 \cdot 10^6} = 1.03 \mu\text{H}$$

$$C_2 = \frac{1}{b_2 \omega_c R} = \frac{1}{1.8478 \cdot 2\pi \cdot 10 \cdot 10^6 \cdot 50} = 172 \text{ pF}$$

$$L_3 = \frac{R}{b_3 \omega_c} = \frac{50}{1.8478 \cdot 2\pi \cdot 10 \cdot 10^6} = 430 \text{ nH}$$

$$C_4 = \frac{1}{b_4 \omega_c R} = \frac{1}{0.7654 \cdot 2\pi \cdot 10 \cdot 10^6 \cdot 50} = 415 \text{ pF}$$

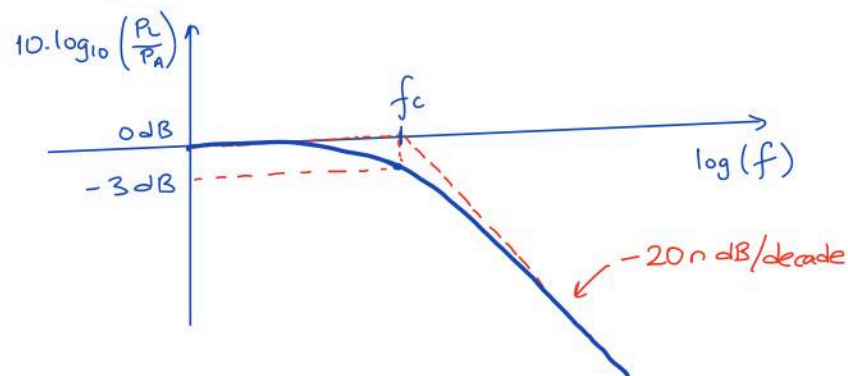
Butterworth LPF :



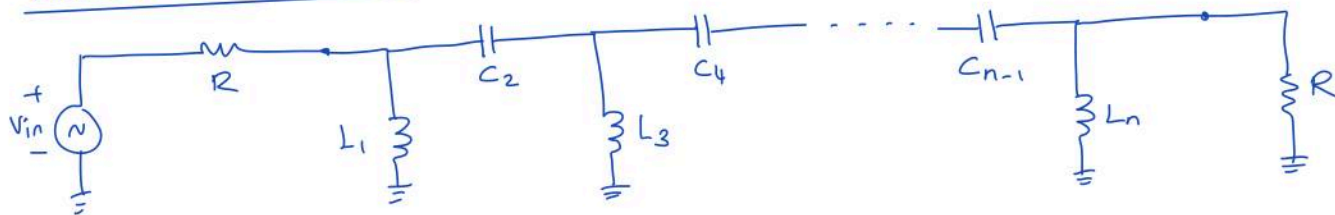
$$C_i = \frac{b_i}{\omega_c R} = \frac{b_i}{2\pi f_c R}$$

$$L_i = \frac{b_i \cdot R}{\omega_c} = \frac{b_i \cdot R}{2\pi f_c}$$

$$\frac{P_L}{P_A} = \frac{1}{1 + \left(\frac{f}{f_c}\right)^{2n}}$$



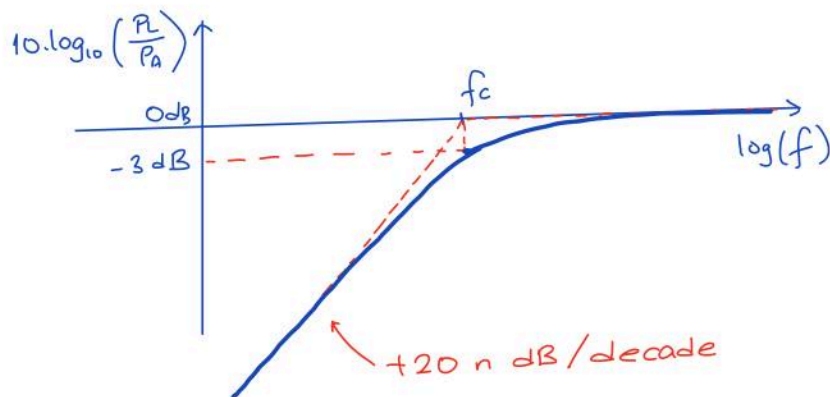
Butterworth HPF:



$$L_i = \frac{R}{b_i \cdot \omega_c} = \frac{R}{b_i \cdot 2\pi f_c}$$

$$C_i = \frac{1}{b_i \omega_c R} = \frac{1}{b_i \cdot 2\pi f_c \cdot R}$$

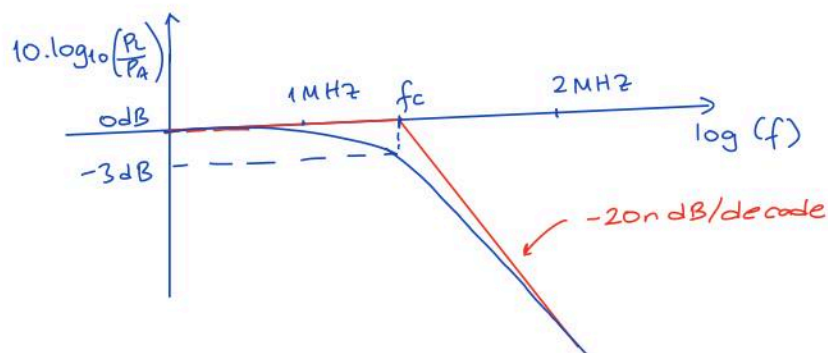
$$\frac{P_L}{P_A} = \frac{\left(\frac{f}{f_c}\right)^{2n}}{1 + \left(\frac{f}{f_c}\right)^{2n}}$$



How do we determine n & f_c ?

Compare the polynomial filter response with the requirements given.

Example: We need a LPF with at most 0.5 dB attenuation at 1 MHz, and at least 30 dB attenuation at 2 MHz.
What is f_c ? What is n ?



From the requirements:

$$1 \text{ MHz} < f_c < 2 \text{ MHz}$$

Question is asking for the minimum n that satisfies the requirements.

* at 1 MHz: $10 \log_{10} \left(\frac{P_L}{P_A} \right) \bigg|_{f=1 \text{ MHz}} = -0.5 \text{ dB}$

$$10 \log_{10} \left(\frac{1}{1 + \left(\frac{10^6}{f_c}\right)^{2n}} \right) = -0.5$$

$$-10 \cdot \log_{10} \left(1 + \left(\frac{10^6}{f_c} \right)^{2n} \right) = -0.5$$

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$$1 + \left(\frac{10^6}{f_c} \right)^{2n} = 10^{0.05}$$

$$\left(\frac{10^6}{f_c} \right)^{2n} = 10^{0.05} - 1 \approx 0.122$$

$$\log_{10} \left(\left(\frac{10^6}{f_c} \right)^{2n} \right) = \log_{10}(0.122)$$

$$2n \cdot \log_{10}(10^6) - 2n \log_{10}(f_c) = \log_{10}(0.122)$$

$$\boxed{12n - 2n \cdot \log_{10}(f_c) = -0.913} \quad (\text{Eq. 1})$$

* At 2 MHz: $10 \cdot \log_{10} \left(\frac{P_L}{P_A} \right) \Big|_{2\text{MHz}} = -30 \text{ dB}$

$$10 \cdot \log_{10} \left(\frac{1}{1 + \left(\frac{2 \cdot 10^6}{f_c} \right)^{2n}} \right) = -30$$

$$\left(\frac{2 \cdot 10^6}{f_c} \right)^{2n} = 10^3 - 1 = 999$$

$$2n \cdot \log_{10}(2 \cdot 10^6) - 2n \cdot \log_{10}(f_c) = \log_{10}(999)$$

$$\boxed{12.6n - 2n \cdot \log_{10}(f_c) = 2.999} \quad (\text{Eq. 2})$$

* 2 equations, 2 unknowns. solve for n & f_c :

Subtracting Eq. 1 from Eq. 2:

$$0.6n = 2.999 - (-0.913) = 3.912$$

$$n = 6.5 \quad \leftarrow \text{minimum } n \text{ that satisfies the requirements.}$$

n must be an integer

so, choose $\boxed{n=7}$

* Now, solve for f_c :

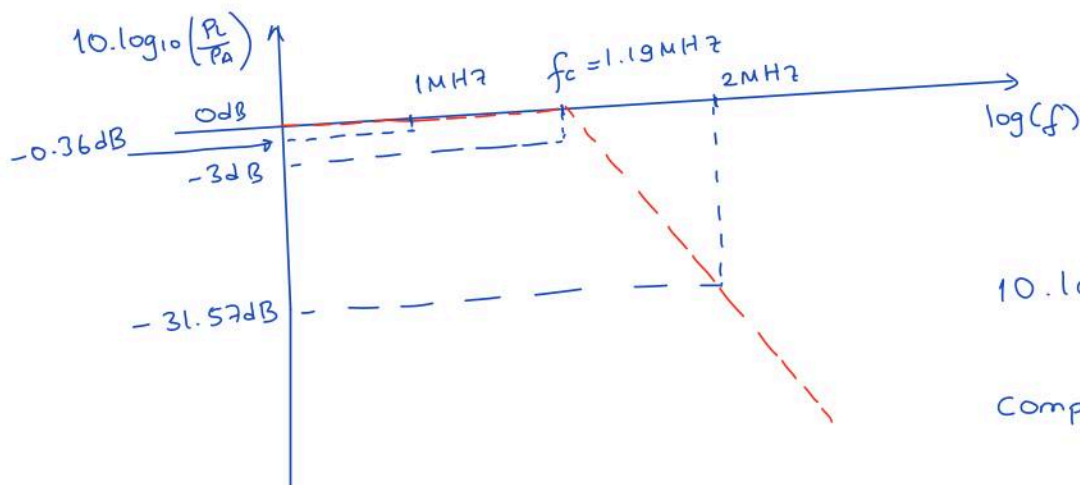
$$\text{Eq. 1: } 12 \times 7 - 14 \log_{10}(f_c) = -0.913 \Rightarrow f_{c,1} = 1.16 \text{ MHz}$$

$$\text{Eq. 2: } 12.6 \times 7 - 14 \log_{10}(f_c) = 2.999 \Rightarrow f_{c,2} = 1.22 \text{ MHz}$$

Any f_c in between would work

Choose f_c as the arithmetic mean: $f_c = \frac{f_{c,1} + f_{c,2}}{2}$

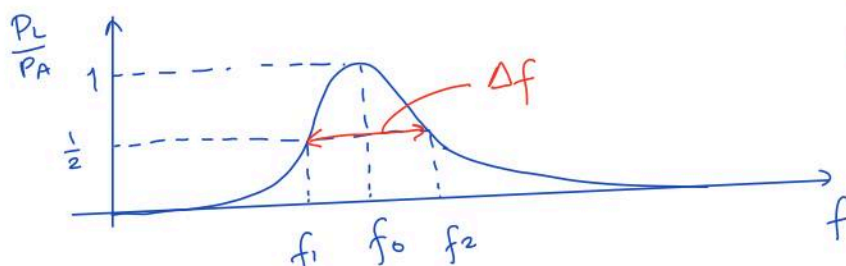
$$f_c = 1.19 \text{ MHz}$$



$$10 \cdot \log_{10} \left(\frac{1}{1 + \left(\frac{f}{f_c} \right)^{2n}} \right)$$

Compute at 1MHz & 2MHz.

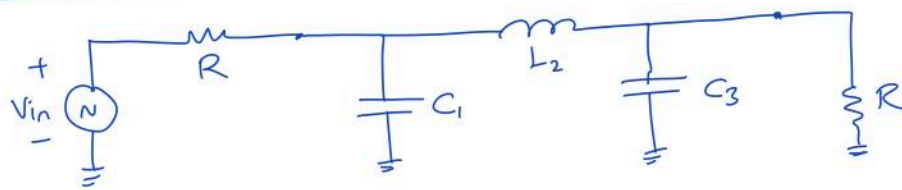
Designing BandPass Filters (BPF):



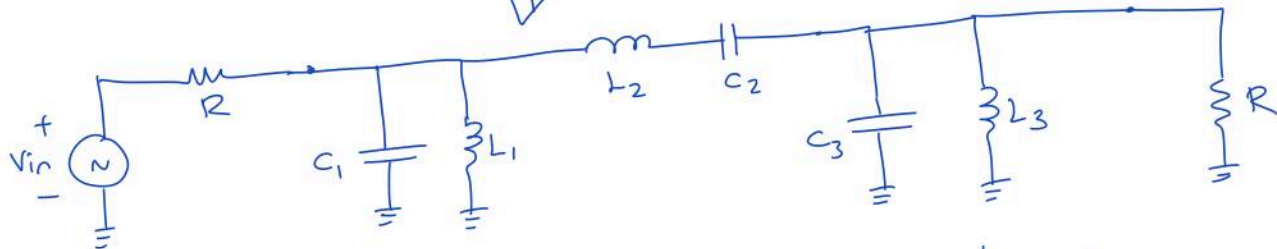
f_0 : resonance frequency
 Δf : 3-dB bandwidth

- 1) Design a LPF with 3-dB cut-off freq f_c equal to Δf of the BPF.
- 2) Then, replace every series and parallel element with a tuned LC circuit at f_0 .

Example : LPF with $f_c = \Delta f$



replace with tuned circuit at f



$$\omega_0 = \frac{1}{\sqrt{L_1 C_1}} = \frac{1}{\sqrt{L_2 C_2}} = \frac{1}{\sqrt{L_3 C_3}}$$

* For Butterworth BPF :

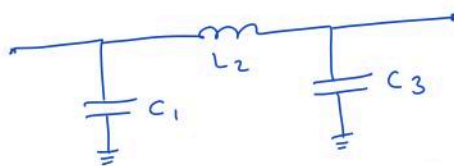
$$\frac{P_L}{P_A} = \frac{1}{1 + \left(\frac{f_0}{\Delta f}\right)^{2n} \left(\frac{f}{f_0} - \frac{f_0}{f}\right)^{2n}}$$

Example : Design a BPF at $f_0 = 1 \text{ GHz}$ and $\text{BW} = 200 \text{ MHz}$.
Assume $n=3$, $R_s = R_L = 50 \Omega$.

$$Q = \frac{f_0}{\Delta f} = \frac{1 \text{ GHz}}{200 \text{ MHz}} = \frac{10^9}{200 \cdot 10^6} = 5$$

Polynomial coefficients : $b_1=1$, $b_2=2$, $b_3=1$

1) Design LPF with $f_c = \Delta f = 200 \text{ MHz}$:



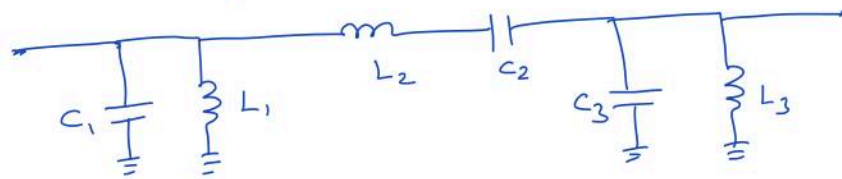
$$C_1 = \frac{b_1}{\omega_c R} = \frac{1}{2\pi \cdot 200 \cdot 10^6 \cdot 50} = 32 \text{ pF}$$

$$L_2 = \frac{b_2 \cdot R}{2\pi f_c} = \frac{2 \cdot 50}{2\pi \cdot 200 \cdot 10^6} = 80 \text{ nH}$$

$$C_3 = C_1 = 32 \text{ pF}$$

2) Tune at $f_0 = 1 \text{ GHz}$:

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$$L_1 = \frac{1}{\omega_0^2 C_1} = 0.8 \text{ nH}$$

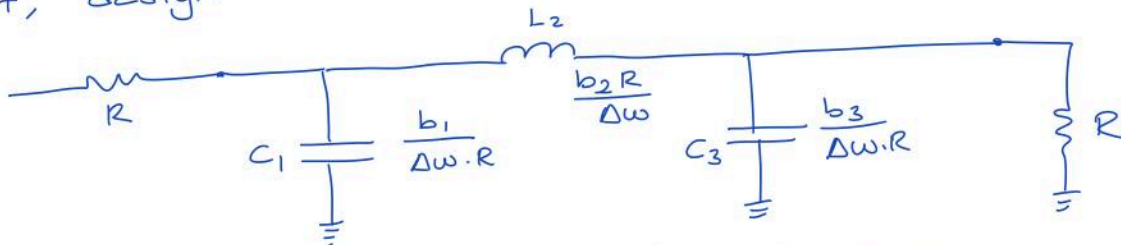
$$C_2 = \frac{1}{\omega_0^2 L_2} = 0.32 \text{ pF}$$

$$L_3 = \frac{1}{\omega_0^2 C_3} = 0.8 \text{ nH}$$

About bandpass filter design

Why do we first design a LPF then tune it?

* First, design a LPF with $\omega_c = \Delta\omega$:

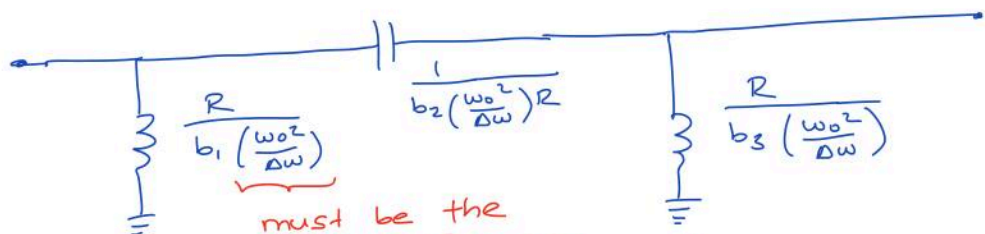


* Then, we add tuned circuits at ω_0 :

$$L_i = \frac{1}{\omega_0^2 C_i} = \frac{1}{\omega_0^2 \frac{b_i}{\Delta\omega \cdot R}} = \frac{R}{b_i \cdot \left(\frac{\omega_0^2}{\Delta\omega}\right)}$$

$$C_i = \frac{1}{\omega_0^2 L_i} = \frac{1}{\omega_0^2 \frac{R}{b_i \cdot \left(\frac{\omega_0^2}{\Delta\omega}\right)}} = \frac{1}{b_i \cdot \left(\frac{\omega_0^2}{\Delta\omega}\right)} \cdot R$$

The components we are adding are:



must be the cut-off frequency of the added HPF !!

We have added a HPF with $\omega_c = \frac{\omega_0^2}{\Delta\omega}$

* so, a BPF is like a combination of a LPF and HPF, with resonance at ω_0 .

$$\frac{P_L}{P_A} = \frac{1}{1 + \left(\frac{f_0}{\Delta f}\right)^{2n} \left(\frac{f}{f_0} - \frac{f_0}{f}\right)^{2n}}$$

* if $\omega \gg \omega_0$, $\frac{P_L}{P_A} \approx \frac{1}{1 + \left(\frac{f_0}{\Delta f}\right)^{2n} \left(\frac{f}{f_0}\right)^{2n}} = \frac{1}{1 + \left(\frac{f}{\Delta f}\right)^{2n}}$

LPF with $f_c = \Delta f$

* if $\omega \ll \omega_0$, $\frac{P_L}{P_A} \approx \frac{1}{1 + \left(\frac{f_0}{\Delta f}\right)^{2n} \left(\frac{f_0}{f}\right)^{2n}} = \frac{1}{1 + \left(\frac{f_0^2/\Delta f}{f}\right)^{2n}}$

$$= \frac{\left(\frac{f}{f_0^2/\Delta f}\right)^{2n}}{1 + \left(\frac{f}{f_0^2/\Delta f}\right)^{2n}}$$

HPF with $f_c = \frac{f_0^2}{\Delta f}$

How to plot :

* 3-dB frequencies at $\frac{P_L}{P_A} = \frac{1}{2}$

$$\frac{f_0}{\Delta f} \cdot \left(\frac{f}{f_0} - \frac{f_0}{f}\right) = \pm 1$$

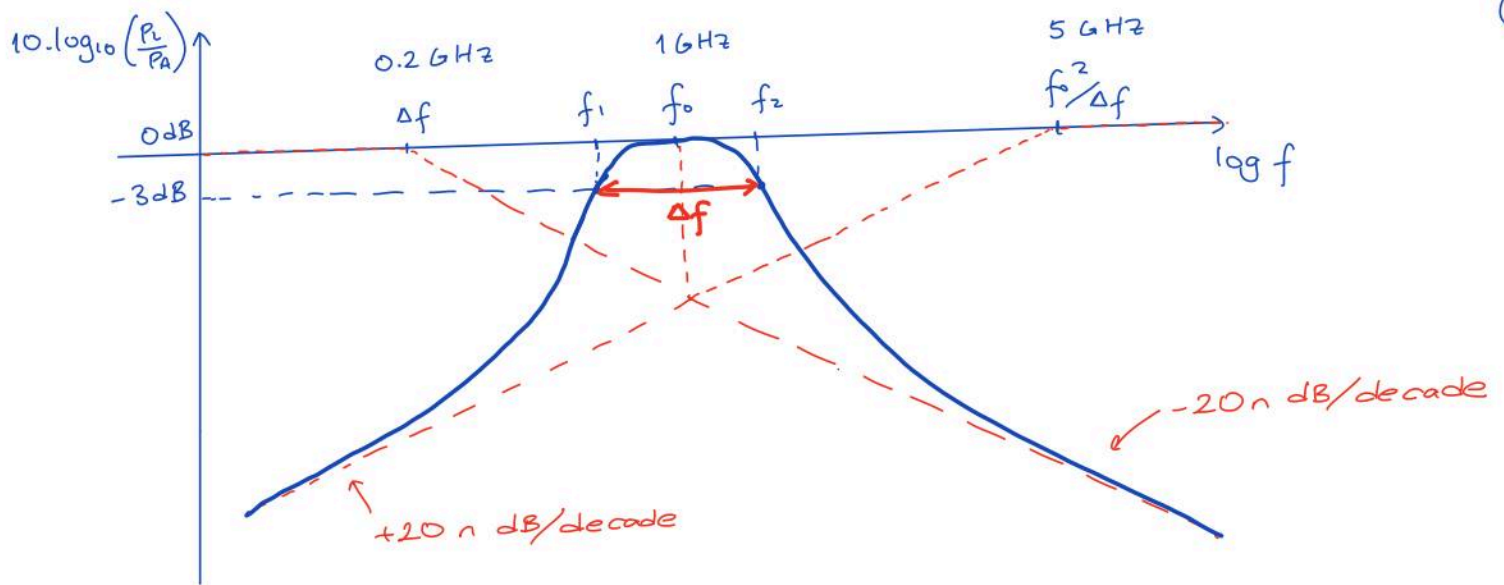
$$\Rightarrow f_{1,2} = \sqrt{f_0^2 + \frac{\Delta f^2}{4}} \pm \frac{\Delta f}{2} = f_0 \sqrt{1 + \left(\frac{\Delta f}{f_0}\right)^2 \cdot \frac{1}{4}} \pm \frac{\Delta f}{2}$$

$$= f_0 \sqrt{1 + \frac{1}{4Q^2}} \pm \frac{\Delta f}{2}$$

$$f_2 - f_1 = \Delta f$$

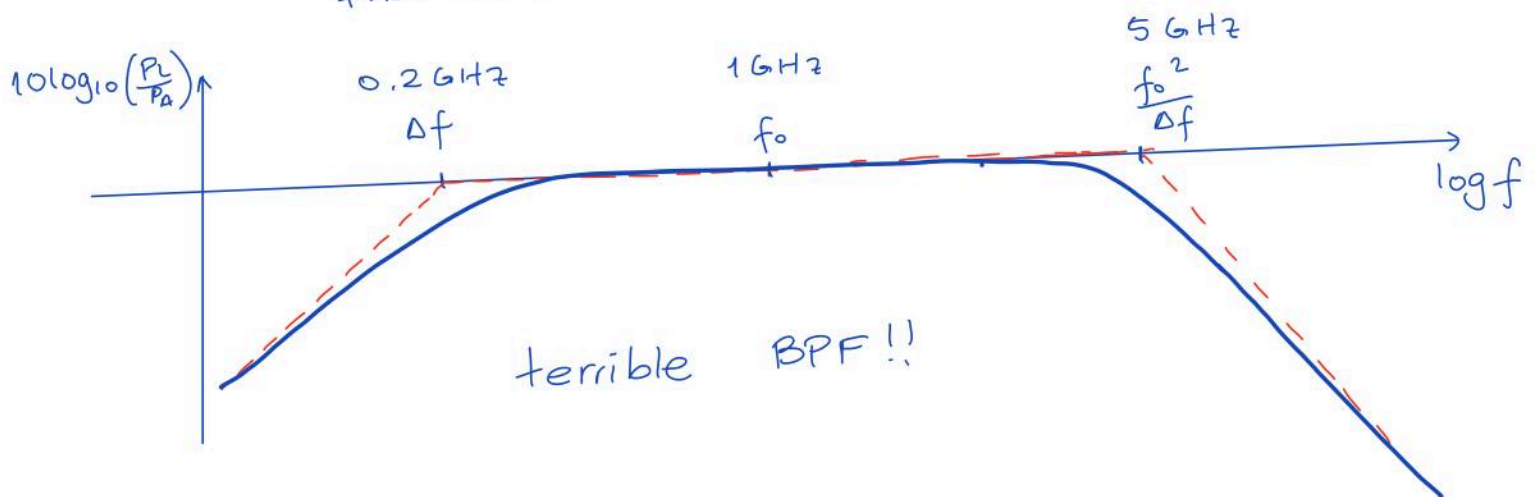
$$f_1 \cdot f_2 = f_0^2$$

* If $Q \geq 3.5$, $f_{1,2} \approx f_0 \pm \frac{\Delta f}{2}$

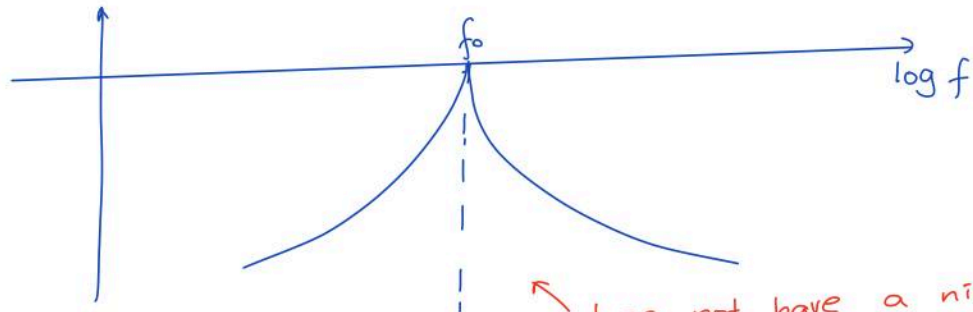
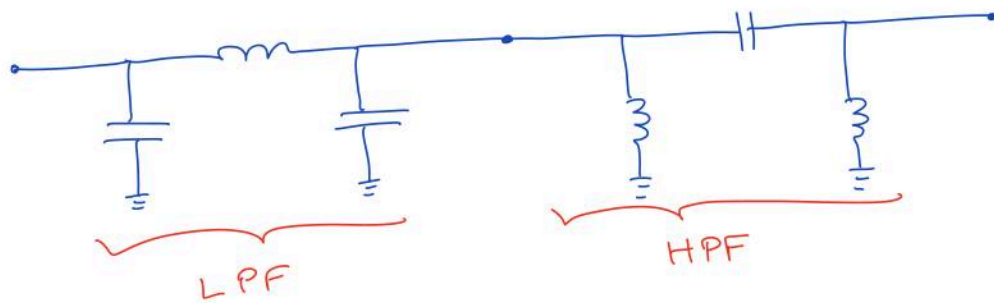


- * Plot the points $(f_1, -3 \text{ dB})$, $(f_0, 0 \text{ dB})$, $(f_2, -3 \text{ dB})$
- * Draw high-frequency asymptote $\Rightarrow$ a LPF with $f_c = \Delta f$.
- 20 n dB/decade slope, passing through $(\Delta f, 0 \text{ dB})$
- * Draw low-frequency asymptote $\Rightarrow$ a HPF with $f_c = \frac{f_0^2}{\Delta f}$
+ 20 n dB/decade slope, passing through $(\frac{f_0^2}{\Delta f}, 0 \text{ dB})$
- * Join the points and the asymptotes.

Question: What would happen if we accidentally start by designing a HPF with $\omega_c = \Delta \omega$ and then tune it at ω_0 ?

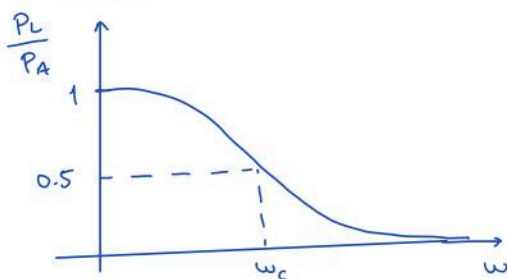


Question: What if we cascade a LPF with a HPF? (17)

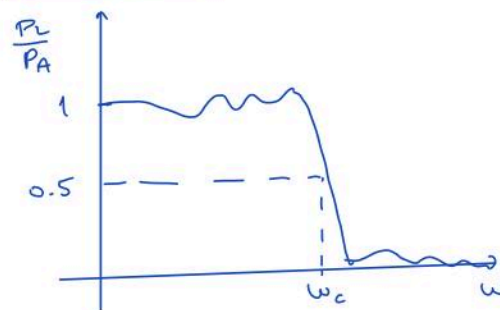


does not have a nice, flat pass band

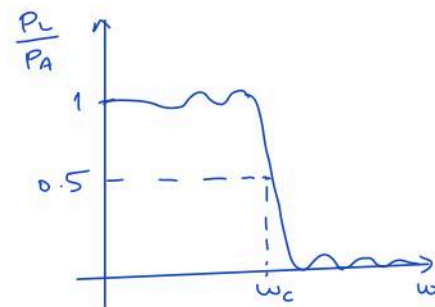
Comparison of Different Polynomial Filters:



Butterworth



Chebyshev



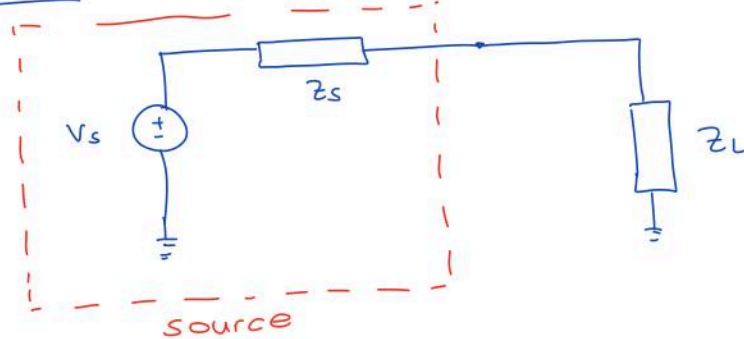
Elliptic

- * Butterworth filter's power transfer function has no ripples.
→ maximally flat (i.e., monotonic) response
- * Trade-off: Butterworth filters have a slower roll-off rate (i.e., slope) at high frequencies.
- * Steps in filter design are identical, only the coefficients in the look-up table are different.

Impedance Matching :

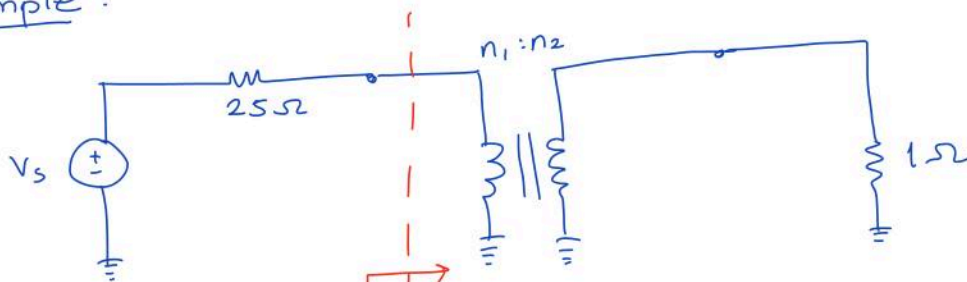
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Remember : For maximum power transfer



$$Z_L = Z_s^*$$

Example :

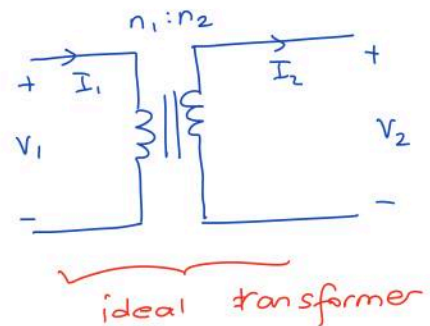
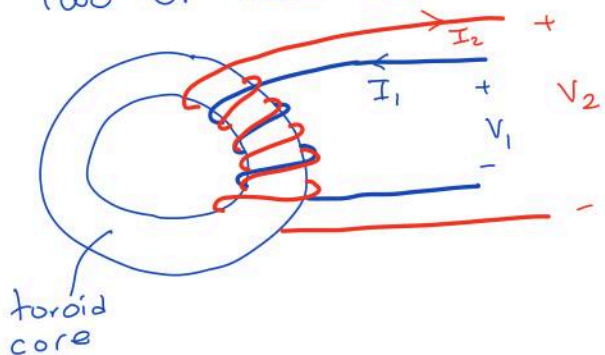


$$R_{in} = \left(\frac{n_1}{n_2}\right)^2 \cdot R_L$$

if we choose $\frac{n_1}{n_2} = 5$, $R_{in} = 25 \Omega$
 $\Rightarrow$ max power delivered to load

Real Transformer :

Two or more coils wound on the same core



* Remember from inductors :

magnetic flux : $\Phi = n A_L \cdot I$

voltage across the inductor : $V = j\omega n \Phi$

- * Here, we have two windings (coupled inductors)
- Both windings are wound in the same "sense".
 - Primary has current flowing into the transformer
 - Secondary " " " out of " "

Total flux: $\Phi = n_1 A_L I_1 - n_2 A_L I_2$

Voltages:
$$\left. \begin{aligned} V_1 &= j\omega n_1 \Phi \\ V_2 &= j\omega n_2 \Phi \end{aligned} \right\} \frac{V_1}{V_2} = \frac{n_1}{n_2}$$

they share
the same total flux

Currents: $\Phi = n_1 A_L I_1 - n_2 A_L I_2$

$$I_1 = \underbrace{\frac{\Phi}{n_1 A_L}}_{\text{new}} + \underbrace{\left(\frac{n_2}{n_1}\right) \cdot I_2}_{\text{what we had for ideal transformer}}$$

⇓

This is called "magnetizing current".
 (We assume this to be zero for ideal transformer.
 → goes to zero for infinitely many turns.

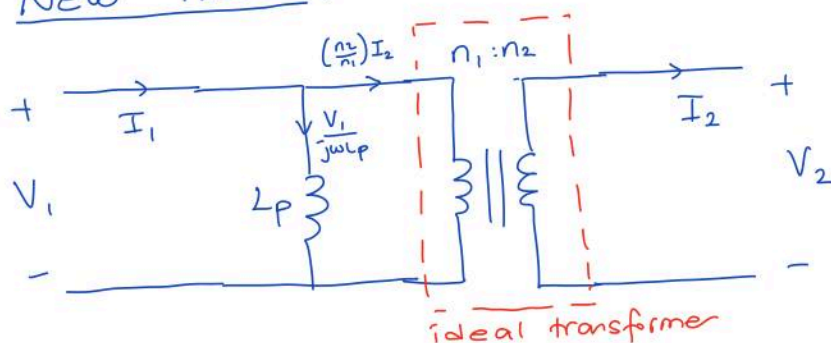
* Substitute: $\Phi = \frac{V_1}{j\omega n_1}$

$$I_1 = \frac{V_1}{j\omega n_1^2 A_L} + \left(\frac{n_2}{n_1}\right) \cdot I_2$$

$L_p = n_1^2 A_L$ (inductance of primary)

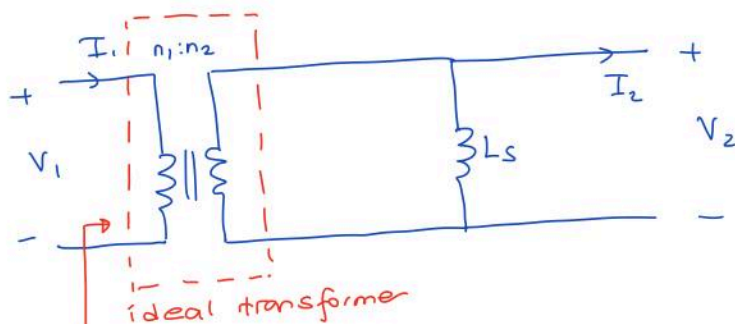
$$I_1 = \frac{V_1}{j\omega L_p} + \left(\frac{n_2}{n_1}\right) \cdot I_2$$

New model:



$$L_p = n_1^2 A_L$$

OR

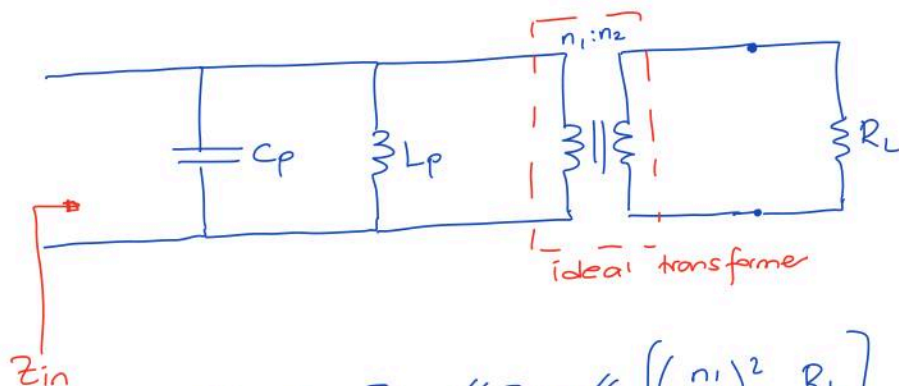


$$L_s = n_2^2 A_L$$

$$Z_{eq} = \left(\frac{n_1}{n_2}\right)^2 j\omega L_s = \left(\frac{n_1}{n_2}\right)^2 \cdot j\omega n_2^2 A_L = j\omega n_1^2 A_L = j\omega L_p$$

Question: How can we cancel out the effect of L_p ?

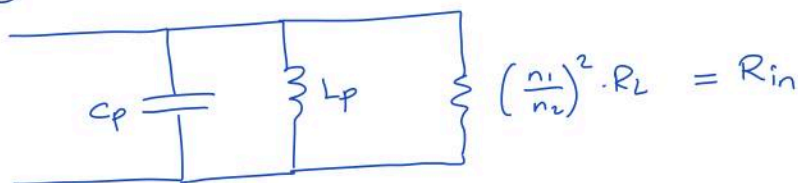
a) Narrowband Transformer: tune out L_p by using a parallel capacitor.
 $\Rightarrow$ only the ideal transformer is left



$$Z_{in} = Z_{Cp} \parallel Z_{Lp} \parallel \left[\left(\frac{n_1}{n_2}\right)^2 \cdot R_L \right]$$

choose $C_p = \frac{1}{\omega_0^2 L_p}$, then, $Z_{in} = \left(\frac{n_1}{n_2}\right)^2 \cdot R_L$

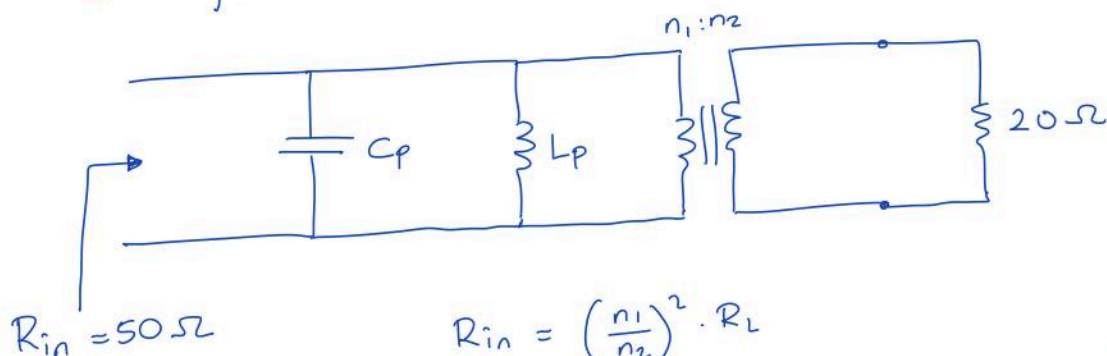
* only works around ω_0 :



$$Q = \frac{R_{in}}{\omega_0 L_p}, \quad \Delta\omega = \frac{\omega_0}{Q}$$

narrowband transformer that works in $\Delta\omega$ bandwidth around ω_0 .

Example: Design a transformer to transform 20Ω into 50Ω at $f_0 = 10\text{ MHz}$. We have $A_L = 3\text{ nH/turns}^2$ core.



$$R_{in} = \left(\frac{n_1}{n_2}\right)^2 \cdot R_L$$

$$50 = \left(\frac{n_1}{n_2}\right)^2 \cdot 20 \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{5}{2}} \approx 1.58$$

Choose $n_1 = 11$, $n_2 = 7 \Rightarrow \frac{n_1}{n_2} \approx 1.57$

Then, $L_p = n_1^2 A_L = 11^2 \cdot 3\text{ nH} = 363\text{ nH}$

$$C_p = \frac{1}{\omega_0^2 L_p} = \frac{1}{(2\pi \cdot 10^7)^2 \cdot 363 \cdot 10^{-9}} = 698\text{ pF}$$

$$* Q = \frac{R_{in}}{\omega_0 L_p} = \frac{50}{2\pi \cdot 10^7 \cdot 363 \cdot 10^{-9}} = 2.2$$

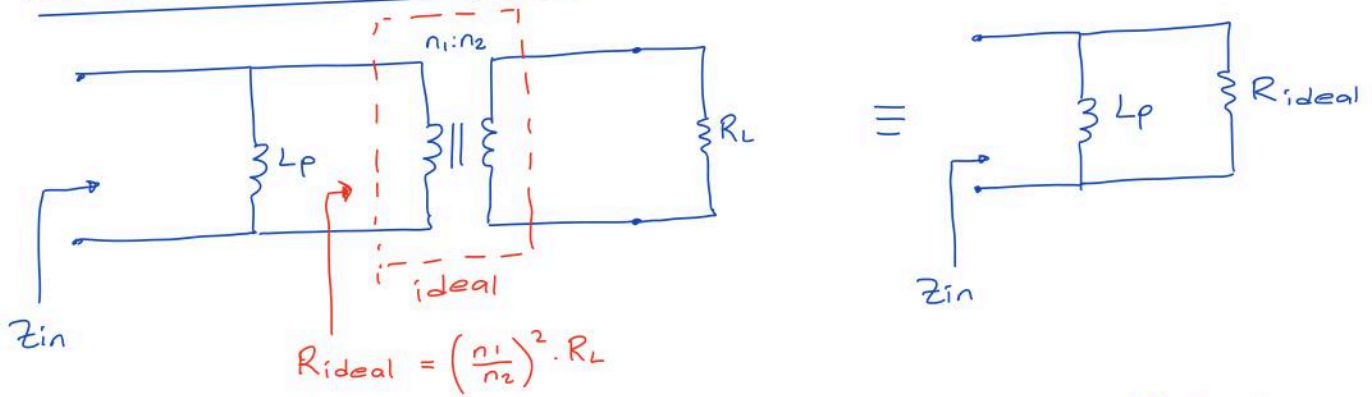
* So, the impedance transformation is valid in a band $\Delta f = f_2 - f_1 = \frac{f_0}{Q} = 4.5\text{ MHz}$

Also,
$$\left. \begin{aligned} f_1 \cdot f_2 &= f_0^2 = 10^2 \\ f_2 - f_1 &= 4.5 \end{aligned} \right\} \begin{array}{l} \text{solve together} \\ f_1 = 8\text{ MHz} \\ f_2 = 12.5\text{ MHz} \end{array}$$

So, the transformation is valid in:

$$8\text{ MHz} \leq f \leq 12.5\text{ MHz}$$

b) Wideband Transformer



$$Z_{in} = j\omega L_p \parallel R_{ideal} = \frac{j\omega L_p \cdot R_{ideal}}{j\omega L_p + R_{ideal}} = \frac{R_{ideal}}{1 + \frac{R_{ideal}}{j\omega L_p}}$$

$$|Z_{in}| = \frac{R_{ideal}}{\sqrt{1 + \left(\frac{R_{ideal}}{\omega L_p}\right)^2}}$$

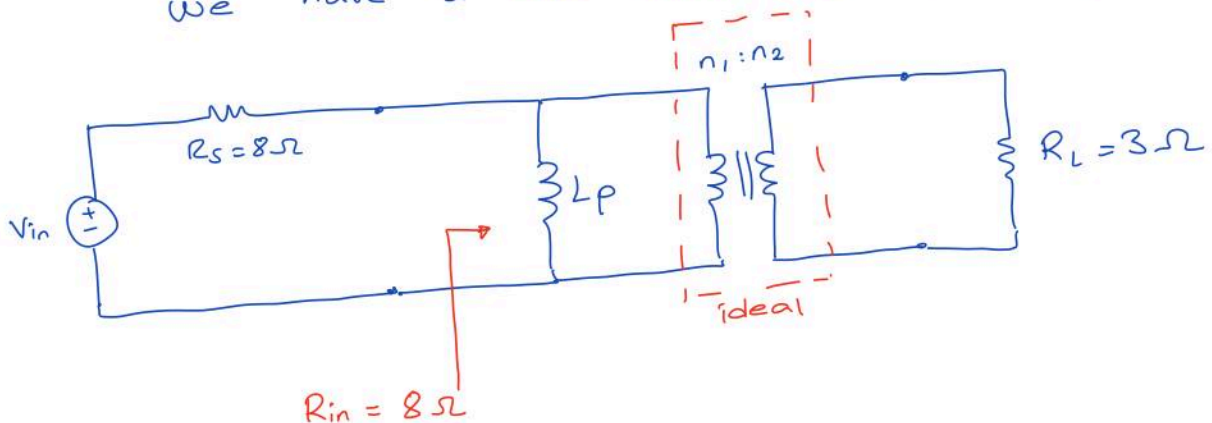
if $\omega L_p = 4 R_{ideal}$
 } denominator ~ 1.03

So, $Z_{in} \approx R_{ideal}$, if $|j\omega L_p| \geq 4 R_{ideal}$

$$L_p \geq \frac{4 R_{ideal}}{\omega_l} = 4 \left(\frac{n_1}{n_2}\right)^2 \cdot \frac{R_L}{\omega_l}$$

→ lowest operating frequency for impedance matching

Example : 1000 W audio amplifier with $R_s = 8 \Omega$, but we have a speaker with $R_L = 3 \Omega$. We need a wideband transformation covering the whole audio range (20 Hz to 20 kHz). We have a core with $A_L = 650 \text{ nH/turns}^2$.



$$R_{in} = \left(\frac{n_1}{n_2}\right)^2 \cdot R_L \Rightarrow 8 = \left(\frac{n_1}{n_2}\right)^2 \cdot 3$$

(23)

$$\frac{n_1}{n_2} = \sqrt{\frac{8}{3}} \approx 1.63$$

To have a wideband transformer:

$$L_p \geq 4 \cdot \frac{R_{in}}{\omega_L} = \frac{4 \cdot 8}{2\pi \cdot 20} \approx 256 \text{ mH}$$

So, using

$$L_p = n_1^2 \cdot A_L \Rightarrow n_1 = \sqrt{\frac{L_p}{A_L}} = \sqrt{\frac{256 \cdot 10^{-3}}{650 \cdot 10^{-9}}} \approx 628 \text{ turns}$$

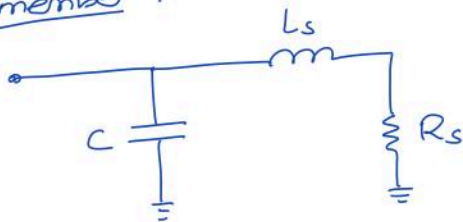
then,

$$n_2 = \frac{n_1}{1.63} \approx 385 \text{ turns}$$

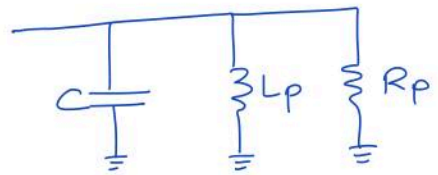
Impedance matching by Resonant Circuits (L-section)

* narrow band impedance matching using RLC circuits.

Remember:



around
 ω_0
 $\equiv$



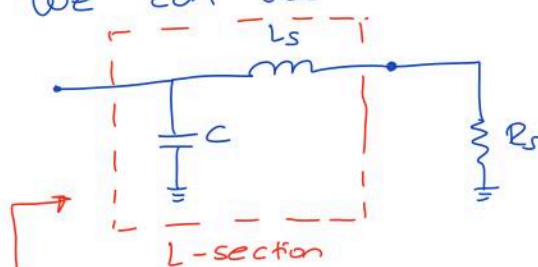
$$\omega_0 = \frac{1}{\sqrt{C L_s \left(1 + \frac{1}{Q^2}\right)}}$$

$$R_p = (Q^2 + 1) R_s$$

$$L_p = \left(1 + \frac{1}{Q^2}\right) L_s$$

$$\omega_0 = \frac{1}{\sqrt{C L_p}}$$

* We can use the same circuit for impedance matching



at ω_0 , $R_p = (Q^2 + 1) R_s$

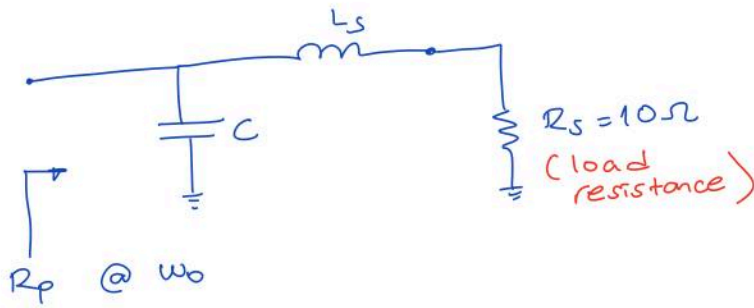
Transforms R_s to R_p .

Valid in a 3-dB bandwidth around ω_0 .

$$\Delta\omega = \frac{\omega_0}{Q}$$

$$Q = \frac{\omega_0 L_s}{R_s} = \frac{R_p}{\omega_0 L_p}$$

Example: Design a resonant matching circuit to convert $10\ \Omega$ to $50\ \Omega$ at 10 MHz . (24)



$$R_p = (Q^2 + 1) R_s$$

$$50 = (Q^2 + 1) 10\ \Omega$$

$$Q = 2$$

$$Q = \frac{\omega_0 L_s}{R_s} = 2 \Rightarrow \frac{2\pi \cdot 10 \cdot 10^6 \cdot L_s}{10} = 2$$

$$L_s = 318\text{ nH}$$

$$\omega_0 = \frac{1}{\sqrt{C \cdot L_s \left(1 + \frac{1}{Q^2}\right)}} \Rightarrow C = \frac{1}{\omega_0^2 L_s \left(1 + \frac{1}{Q^2}\right)}$$

$$= 637\text{ pF}$$

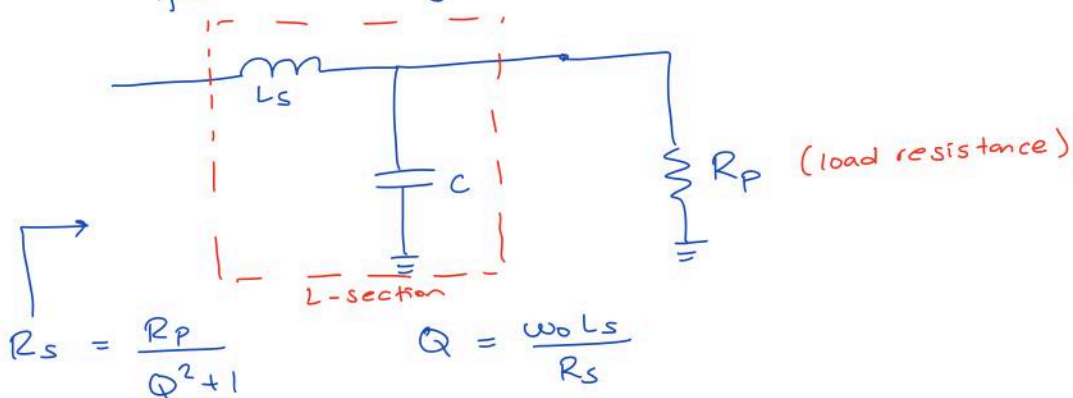
* This transformation is valid in:

$$Q = \frac{\omega_0}{\Delta\omega} = \frac{f_0}{\Delta f} = 2 \Rightarrow \Delta f = 5\text{ MHz}$$

$$\left. \begin{aligned} f_2 - f_1 &= \Delta f = 5 \\ f_1 \cdot f_2 &= f_0^2 = 100 \end{aligned} \right\} \begin{aligned} f_1 &= 7.8\text{ MHz} \\ f_2 &= 12.8\text{ MHz} \end{aligned}$$

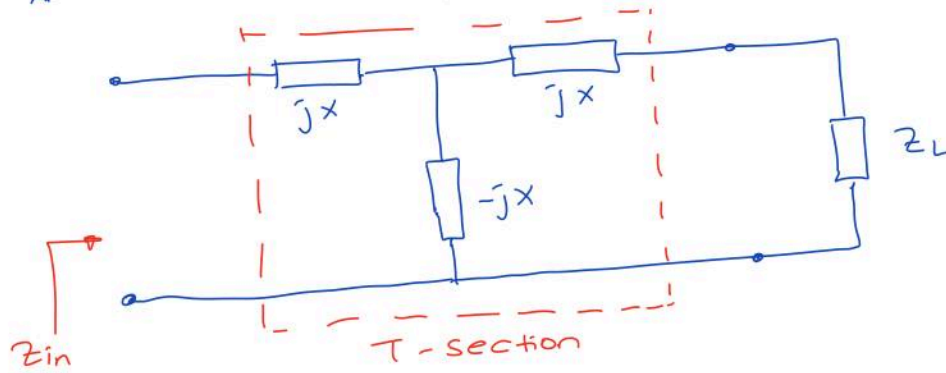
So, valid in: $7.8\text{ MHz} \leq f \leq 12.8\text{ MHz}$

Note: to transform to a smaller resistance from a larger resistance



Impedance Inverter

* narrowband impedance transformation



$$\begin{aligned}
 Z_{in} &= jX + (-jX) \parallel (jX + Z_L) \\
 &= jX + \left(\frac{1}{-jX} + \frac{1}{jX + Z_L} \right)^{-1} = jX + \left(\frac{jX + Z_L - jX}{-jX(jX + Z_L)} \right)^{-1} \\
 &= jX + \frac{(jX + Z_L)(-jX)}{Z_L} = \frac{jXZ_L + X^2 - jXZ_L}{Z_L} = \frac{X^2}{Z_L}
 \end{aligned}$$

$$\boxed{Z_{in} = \frac{X^2}{Z_L}} \quad \left. \vphantom{\frac{X^2}{Z_L}} \right\} Z_L \text{ is inverted and scaled}$$

Example: Design an impedance inverter to convert $Z_L = 10\Omega$ to $Z_{in} = 50\Omega$ at $f = 10 \text{ MHz}$.

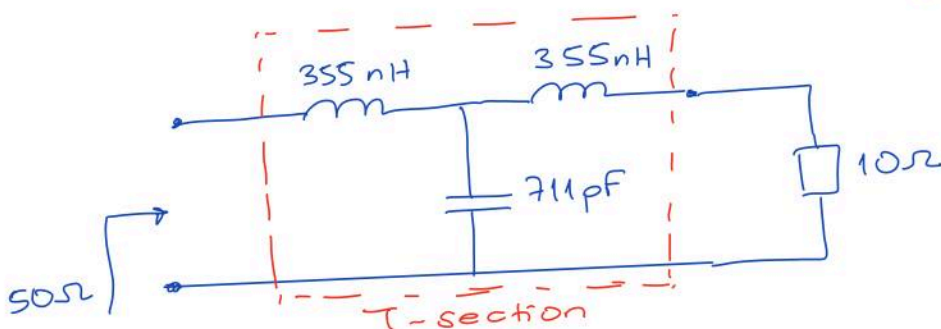
$$Z_{in} = \frac{X^2}{Z_L} \Rightarrow X^2 = Z_{in} \cdot Z_L = 500 \Rightarrow X \approx 22.4 \Omega$$

* for jX elements, use inductors: $jX = j\omega L$

$$L = \frac{X}{\omega} = \frac{22.4}{2\pi \cdot 10 \cdot 10^6} = 355 \text{ nH}$$

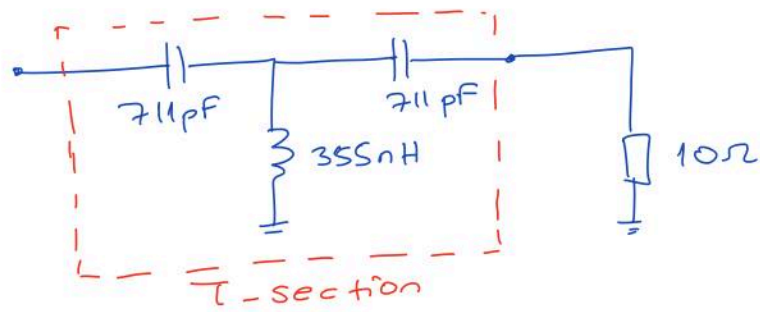
* for $-jX$ element, use a capacitor: $-jX = \frac{1}{j\omega C}$

$$C = \frac{1}{X\omega} = \frac{1}{22.4 \times 2\pi \cdot 10^7} = 711 \text{ pF}$$

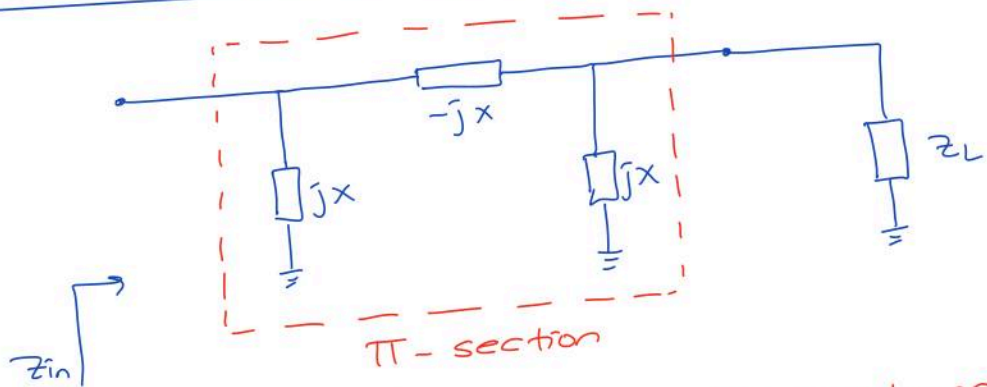


Other solution : $X^2 = 500 \Rightarrow X = -22.4 \Omega$

- * jX components $\Rightarrow$ capacitors : $C = 711 \text{ pF}$
- $-jX$ component $\Rightarrow$ inductor : $L = 355 \text{ nH}$



Another impedance inverter : π -section



$$Z_{in} = \frac{X^2}{Z_L}$$

X : reactance
(could be positive or negative)

Crystal Filters:

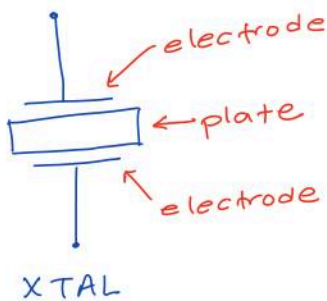
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- LC tuned circuits do not have high Q
(max 100, typically 10)
- mechanical devices typically have higher Q
→ provide much better frequency selectivity

Quartz Crystals:

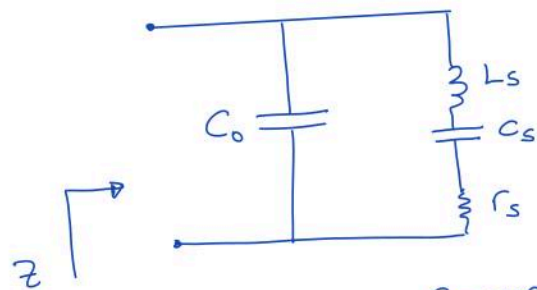
- piezoelectric materials
- convert electrical field into mechanical vibrations at the same frequency as the applied field
- stable, very small loss due to friction → very high Q
- used in timing circuits in watches, computers, etc.
very precise resonance → precise timing

$Q \approx 100,000 !!!$



$\equiv$

Equivalent circuit



- * Plate dimensions define mechanical resonant frequency
- $\left\{ \begin{array}{l} L_s : \text{proportional to the mass of the plate} \\ C_s : \text{determined by the stiffness of the quartz} \\ r_s : \text{models friction losses during vibration} \end{array} \right.$ mechanical
- $\left\{ \begin{array}{l} C_0 : \text{actual capacitance between the electrodes} \end{array} \right.$ electrical

- * Two resonance frequencies:

$$f_s = \frac{1}{2\pi \sqrt{L_s C_s}}$$

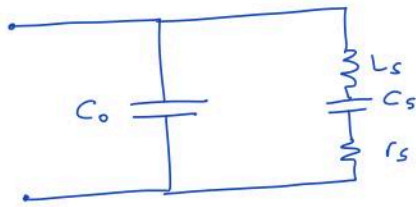
: series resonance. High Q
⇒ all mechanical

$$f_p = \frac{1}{2\pi \sqrt{L_s \cdot \frac{C_s C_0}{C_s + C_0}}}$$

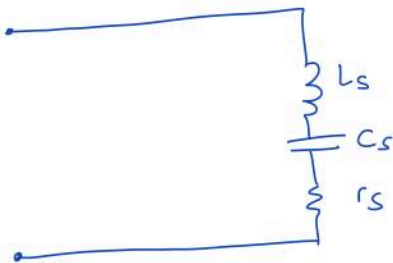
C_s in series with C_0

: parallel resonance. Lower Q
due to actual capacitance C_0 .

* We will use this device at series resonance $\Rightarrow$ high Q
 $L_s \sim 10 \text{ mH}$, $C_s \sim 1 \text{ fF}$ (10^{-15} F)
 $r_s \sim 1-10 \Omega$, $C_o \sim 1 \text{ pF}$ (10^{-12} F)



||| around f_s



Around f_s :

$$|Z_{\omega}| = \left| \frac{1}{j\omega_s C_o} \right| \sim 10^6 \Omega \gg r_s$$

$\downarrow$ MHz $\downarrow$ pF
 $\sim 10^6 \times 10^{-12}$

We can ignore C_o
 when operating around f_s .

In TRC-11 : Quartz crystals are used in:

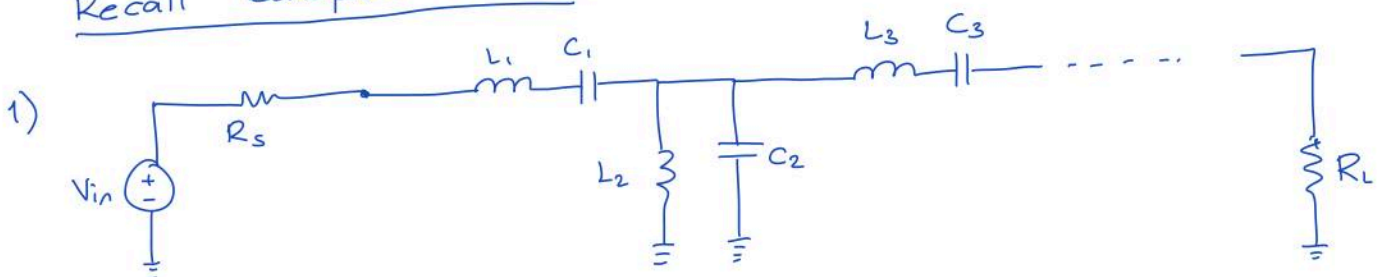
- 1) 15 MHz oscillator, 12 MHz oscillator
- 2) very narrow band IF filter with $f_0 = 15 \text{ MHz}$.

IF Ladder Filter (example for TRC-10)

IF: intermediate frequency (shift signal to the frequency of the quartz crystal to enable high Q filtering.)

- In TRC-10, 16 MHz XTAL has a BW of 100-200 Hz.
- We need $\sim 10 \text{ kHz}$ BW for speech/music signal.
- So, we use crystals in ladder topology to create this BW.

Recall Bandpass Filters:

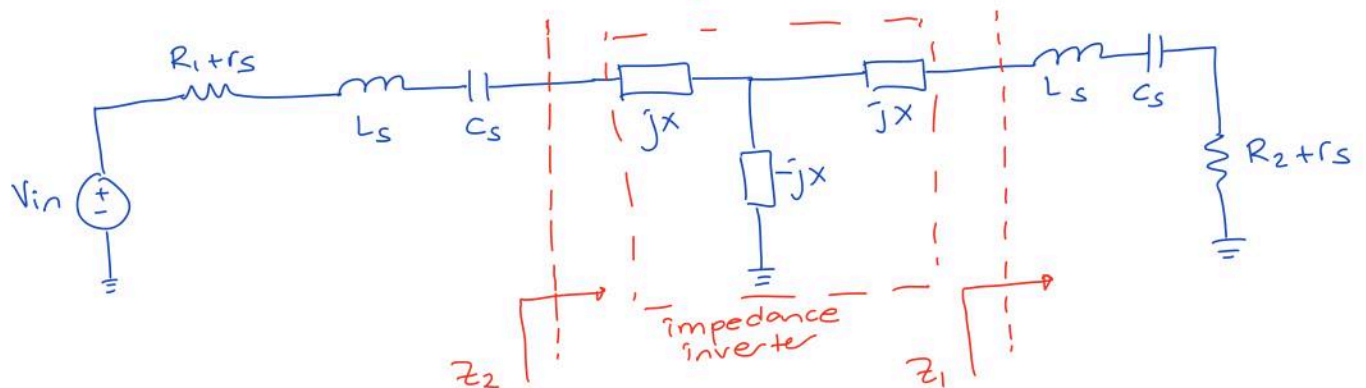
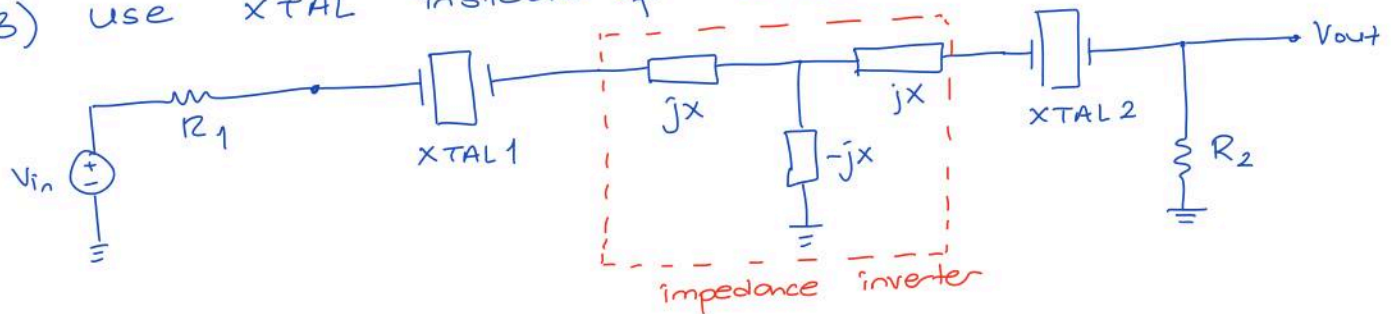


2) Recall crystal equivalent circuit:

(29)



3) Use XTAL instead of series tuned LC

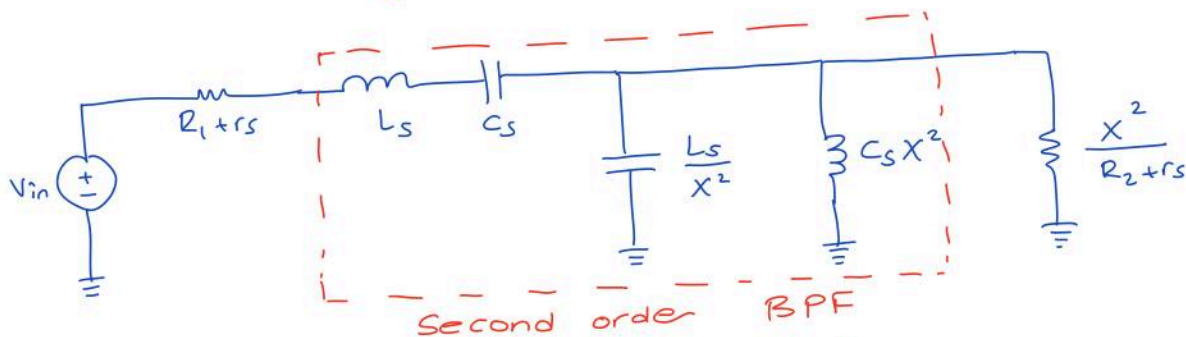


$$Z_1 = j\omega L_s + \frac{1}{j\omega C_s} + (R_2 + r_s)$$

$$Z_2 = \frac{X^2}{Z_1} = \frac{X^2}{j\omega L_s + \frac{1}{j\omega C_s} + (R_2 + r_s)} = \frac{1}{j\omega \frac{L_s}{X^2} + \frac{1}{j\omega C_s X^2} + \left(\frac{R_2 + r_s}{X^2}\right)}$$

$$= \left(\underbrace{j\omega \frac{L_s}{X^2}}_{\text{capacitor, } C = \frac{L_s}{X^2}} + \underbrace{\frac{1}{j\omega C_s X^2}}_{\text{inductor, } L = C_s X^2} + \underbrace{\frac{R_2 + r_s}{X^2}}_{\text{resistor, } R = \frac{X^2}{R_2 + r_s}} \right)^{-1}$$

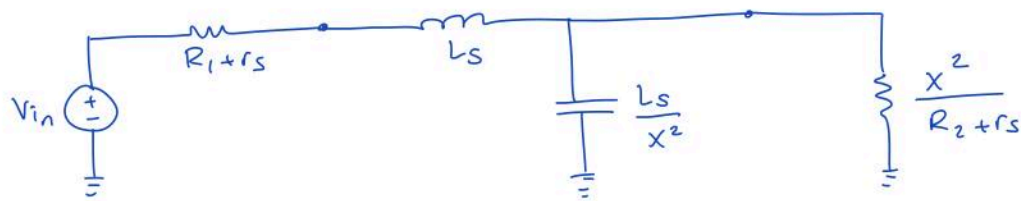
} parallel connection of 3 components



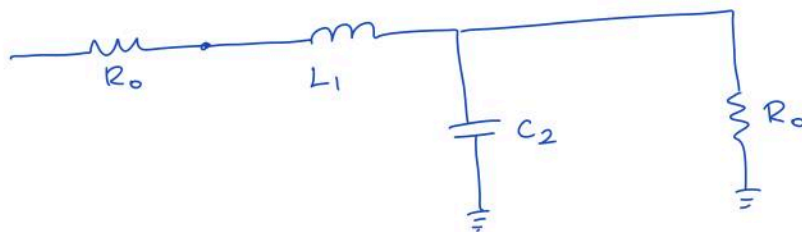
$$\omega_0 = \frac{1}{\sqrt{L_s C_s}} \quad \omega_0 = \frac{1}{\sqrt{\frac{L_s}{X^2} \cdot C_s \cdot X^2}} = \frac{1}{\sqrt{L_s C_s}}$$

Design 2nd order Butterworth BPF

* First, design a 2nd order LPF with $f_c = \Delta f$.
 $b_1 = b_2 = 1.414$



* We want $R_1 + r_s = \frac{X^2}{R_2 + r_s} = R_o$ for maximum power transfer



$$L_1 = \frac{b_1 \cdot R_o}{\Delta \omega} = L_s$$

$$C_2 = \frac{b_2}{\Delta \omega \cdot R_o} = \frac{L_s}{X^2}$$

Divide L_1 by C_2

$$X^2 = R_o^2 \Rightarrow X = R_o = \frac{L_s \cdot \Delta \omega}{b_1}$$

$$X = R_o = \frac{L_s \cdot \Delta \omega}{1.414}$$

$$R_o = R_1 + r_s = R_2 + r_s$$

L_s, C_s, r_s are fixed due to the crystal.
 We need to pick X, R_1, R_2 .

Example: 16 MHz XTAL with $L_s = 15 \text{ mH}$, $r_s = 15 \Omega$.
 We want 10 kHz BW.

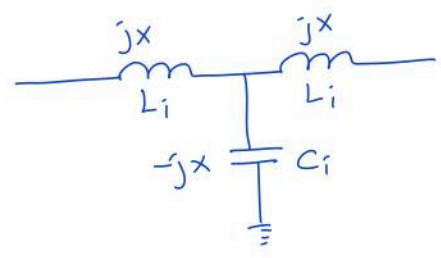
So, for LPF, $f_c = \Delta f = 10 \text{ kHz}$

$$X = R_o = \frac{\Delta \omega \cdot L_s}{1.414} = \frac{2\pi \cdot 10 \cdot 10^3 \cdot 15 \cdot 10^{-3}}{1.414} \approx 666 \Omega$$

$$R_o = R_1 + r_s \Rightarrow R_1 = 651 \Omega$$

$$R_o = \frac{X^2}{R_2 + r_s} = \frac{R_o^2}{R_2 + r_s} \Rightarrow R_o = R_2 + r_s \Rightarrow R_2 = 651 \Omega$$

* Now, design the impedance inverter:



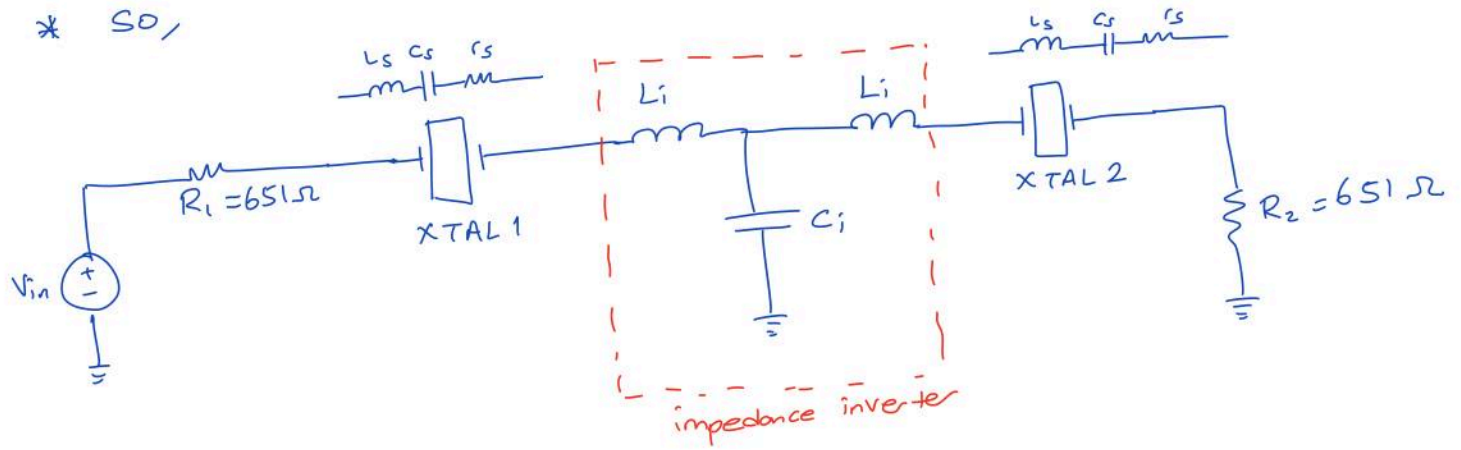
$$jX = j2\pi f_o \cdot L_i$$

$$L_i = \frac{X}{2\pi f_o} = \frac{666}{2\pi \cdot 16 \cdot 10^6} = 6.6 \mu H$$

$$-jX = \frac{1}{j2\pi f_o \cdot C_i}$$

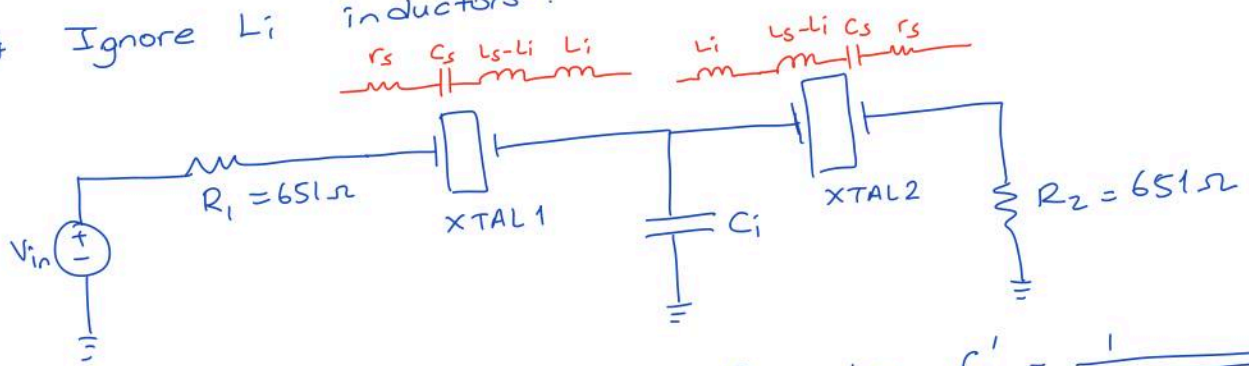
$$C_i = \frac{1}{2\pi f_o \cdot X} = \frac{1}{2\pi \cdot 16 \cdot 10^6 \cdot 666} = 14.9 pF$$

* So,



* Design trick in TRC-10 IF ladder filter:
 $L_i = 6.6 \mu H \ll L_s (\sim 15 mH)$

* Ignore L_i inductors:



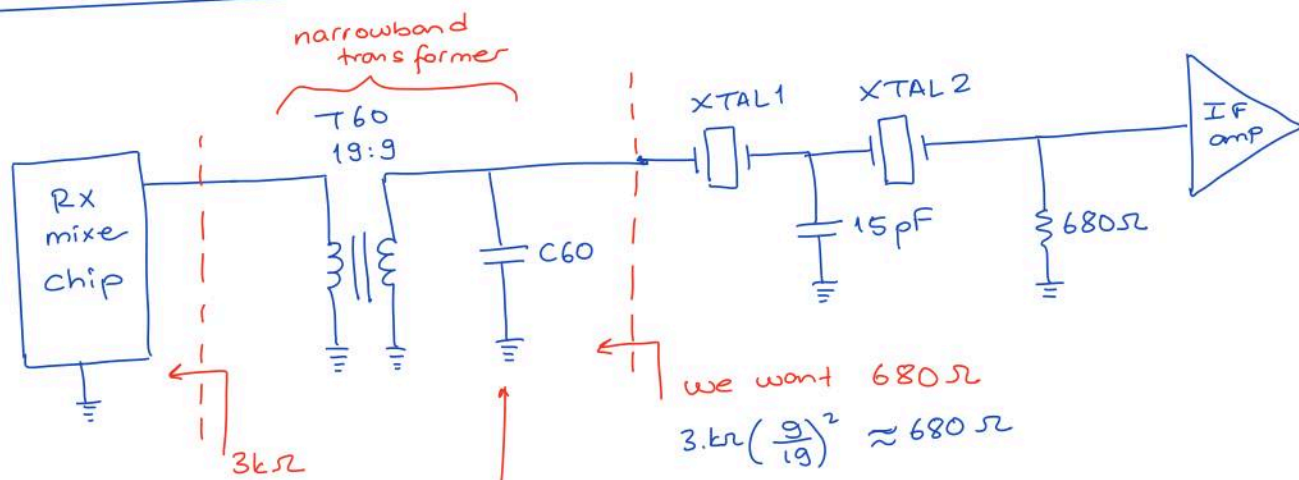
This only slightly changes f_s to $f_s' = \frac{1}{2\pi \sqrt{C_s \cdot (L_s - L_i)}}$

BPF at $f_o = 16 MHz$, with $\Delta f = 10 kHz$

* What is the Q? $Q = \frac{16 MHz}{10 kHz} = \frac{f_o}{\Delta f}$

$$= \frac{16 \cdot 10^6}{10 \cdot 10^3} = 1600 !!! \text{ VERY HIGH}$$

Overall Section in TRC-10



we want 680 Ω
 $3.k\Omega \left(\frac{9}{19}\right)^2 \approx 680 \Omega$

to tune out L_s (inductance of secondary)

For transformer: $A_L = 20 \text{ nH/turns}^2$

$$L_s = n_2^2 A_L = 9^2 \times 20 = 1620 \text{ nH}$$

$$C_{60} = \frac{1}{\omega_0^2 L_s} \approx 60 \text{ pF} \quad (56 \text{ pF in TRC-10})$$